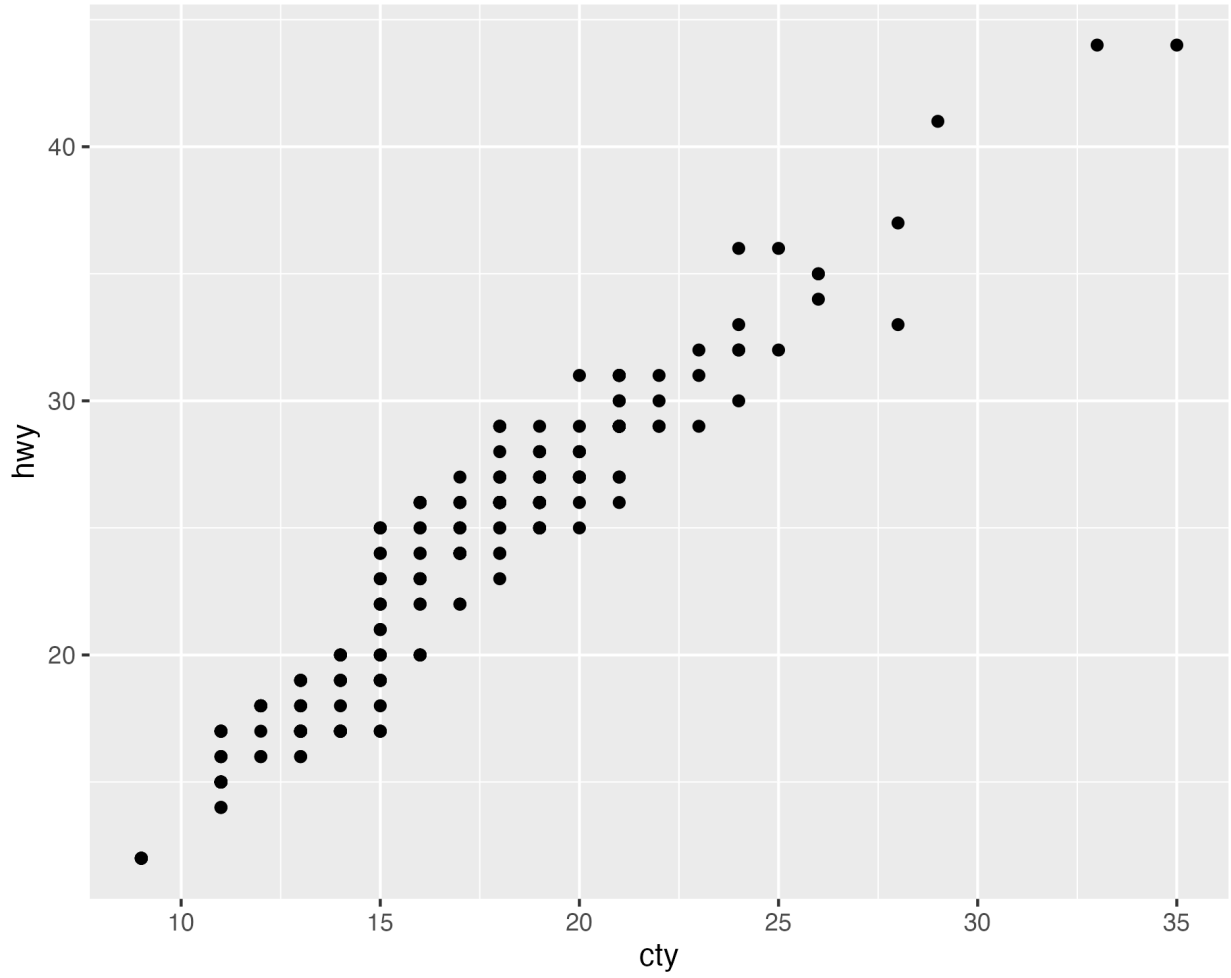
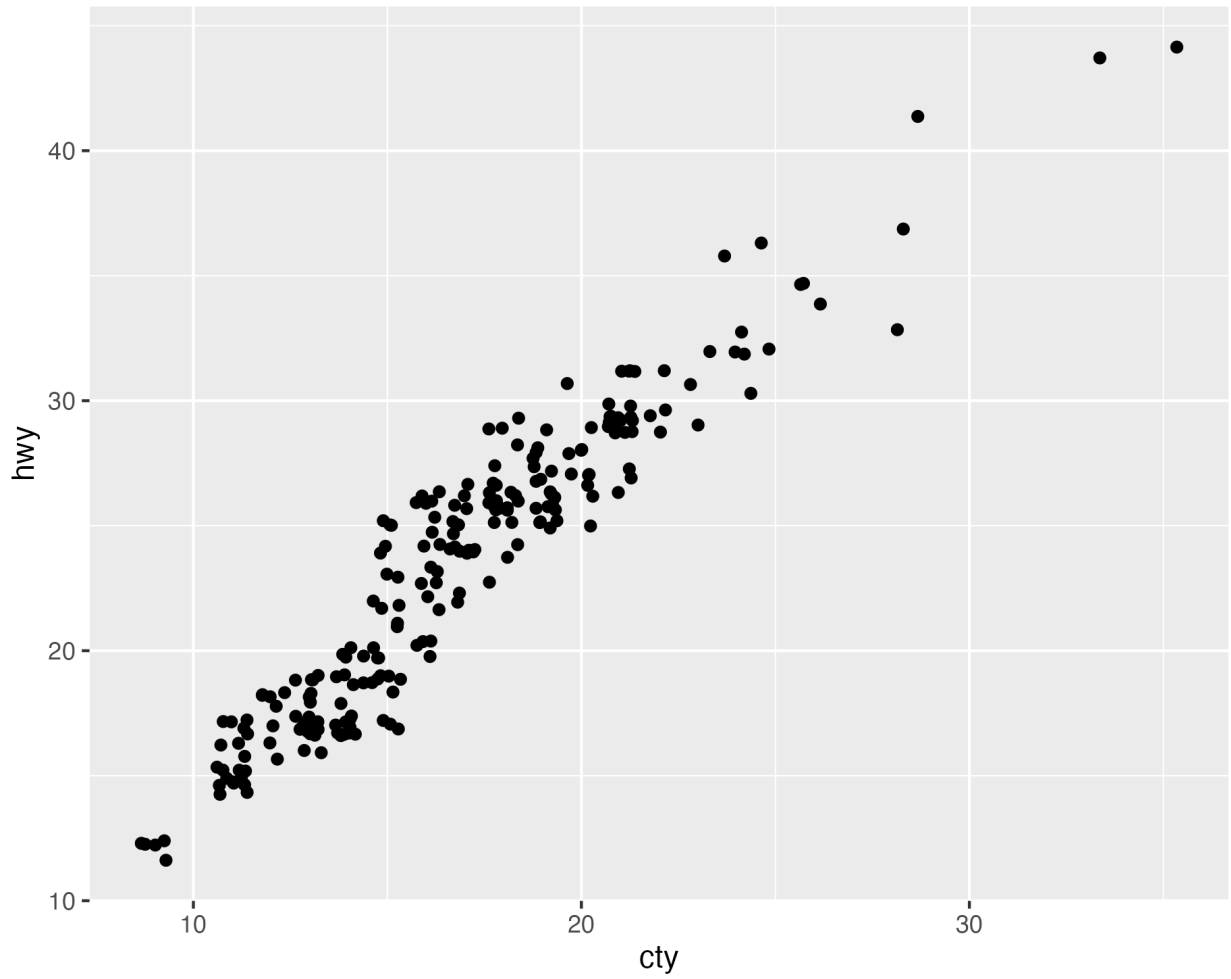
2.3.1 Exercises

ggplot(mpg, aes(cty, hwy)) + geom_point()



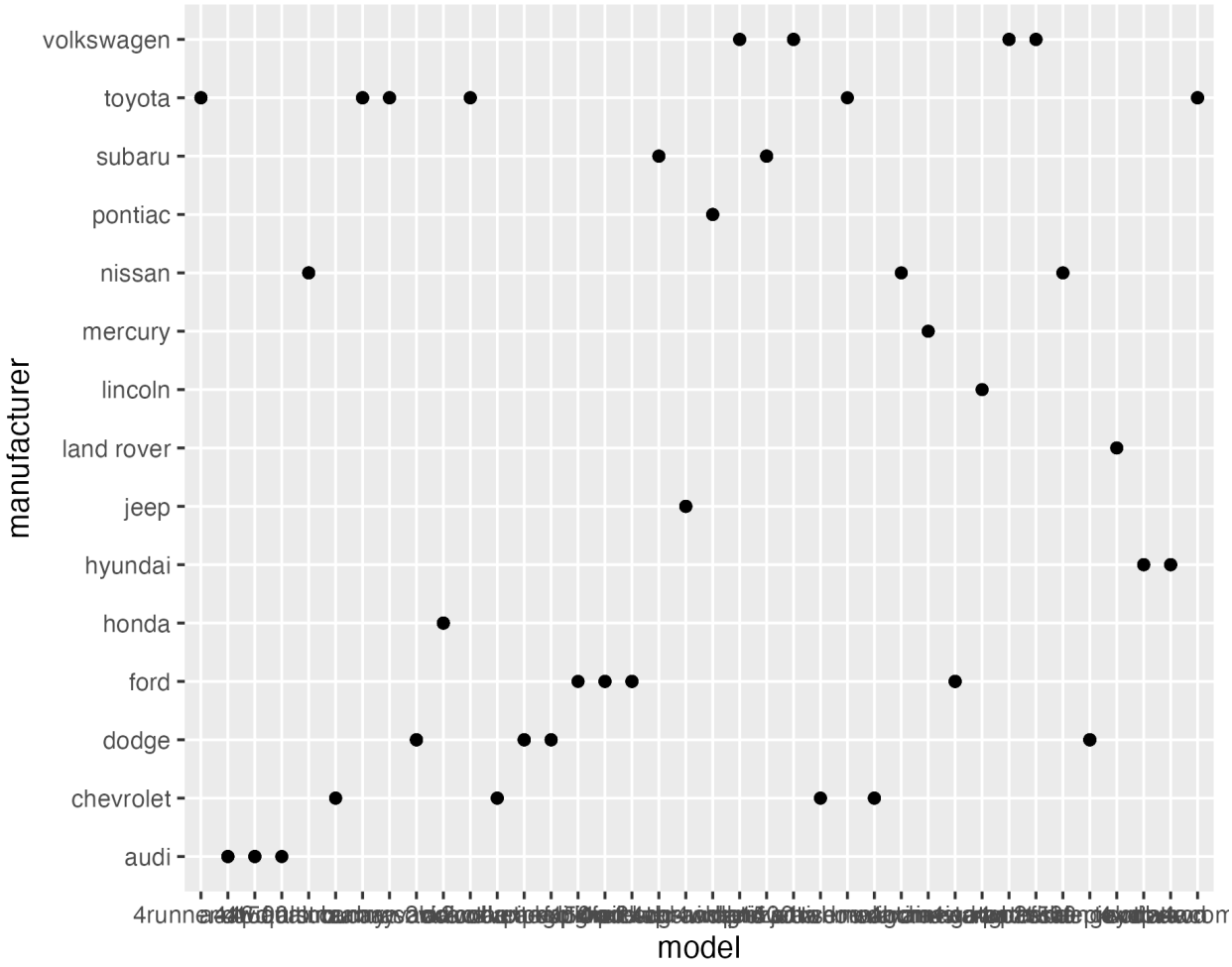
strong linear correlation, makes sense as cars that are fuel efficient in the city are usually fuel efficient on the highways as well

ggplot(mpg, aes(cty, hwy)) + geom_jitter()



using jitter makes it clear that there are quite a few data overlapping in the original scatterplot.
this may prove as problematic when reading and drawing conclusions from the initial graph.

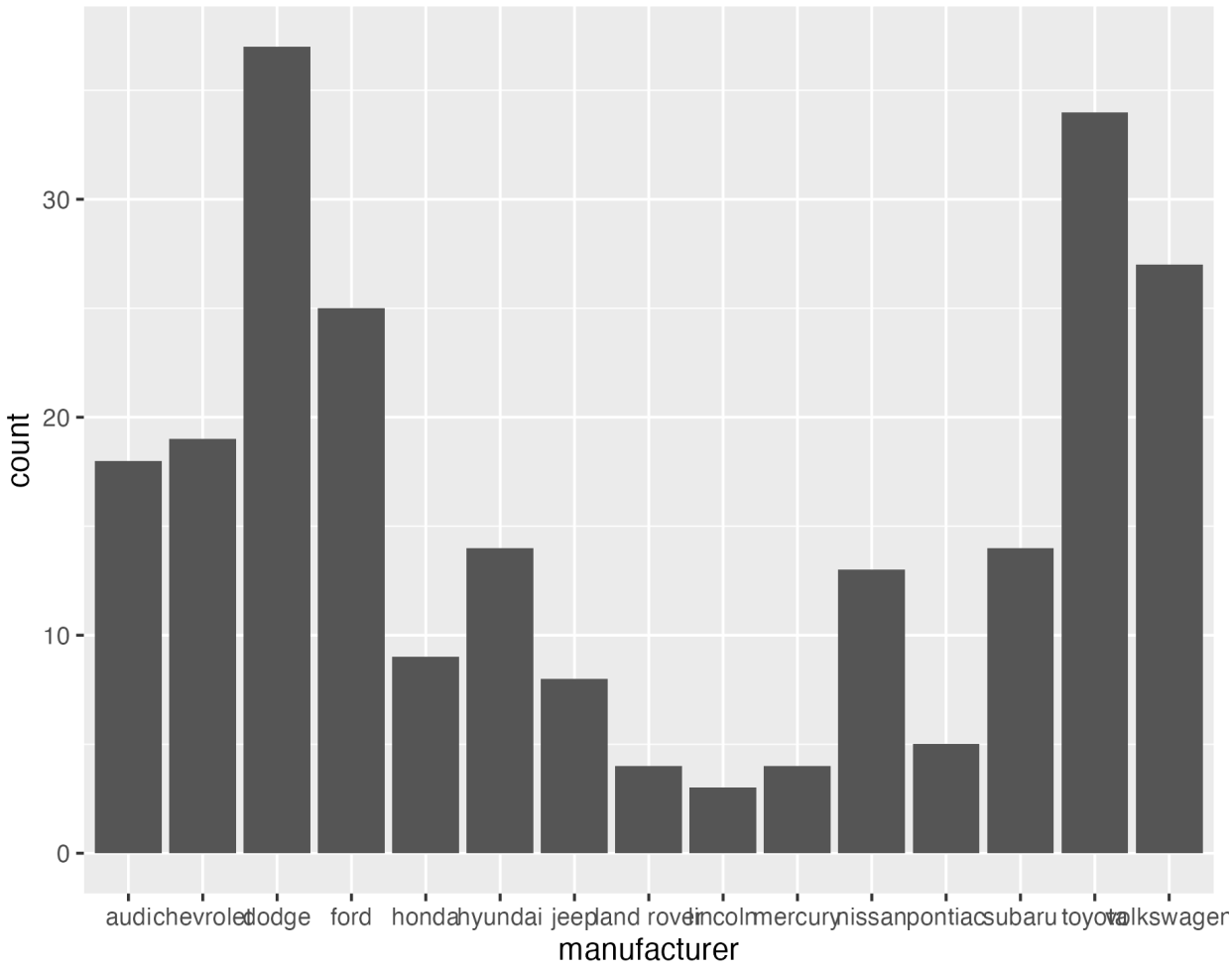
ggplot(mpg, aes(model, manufacturer)) + geom_point()



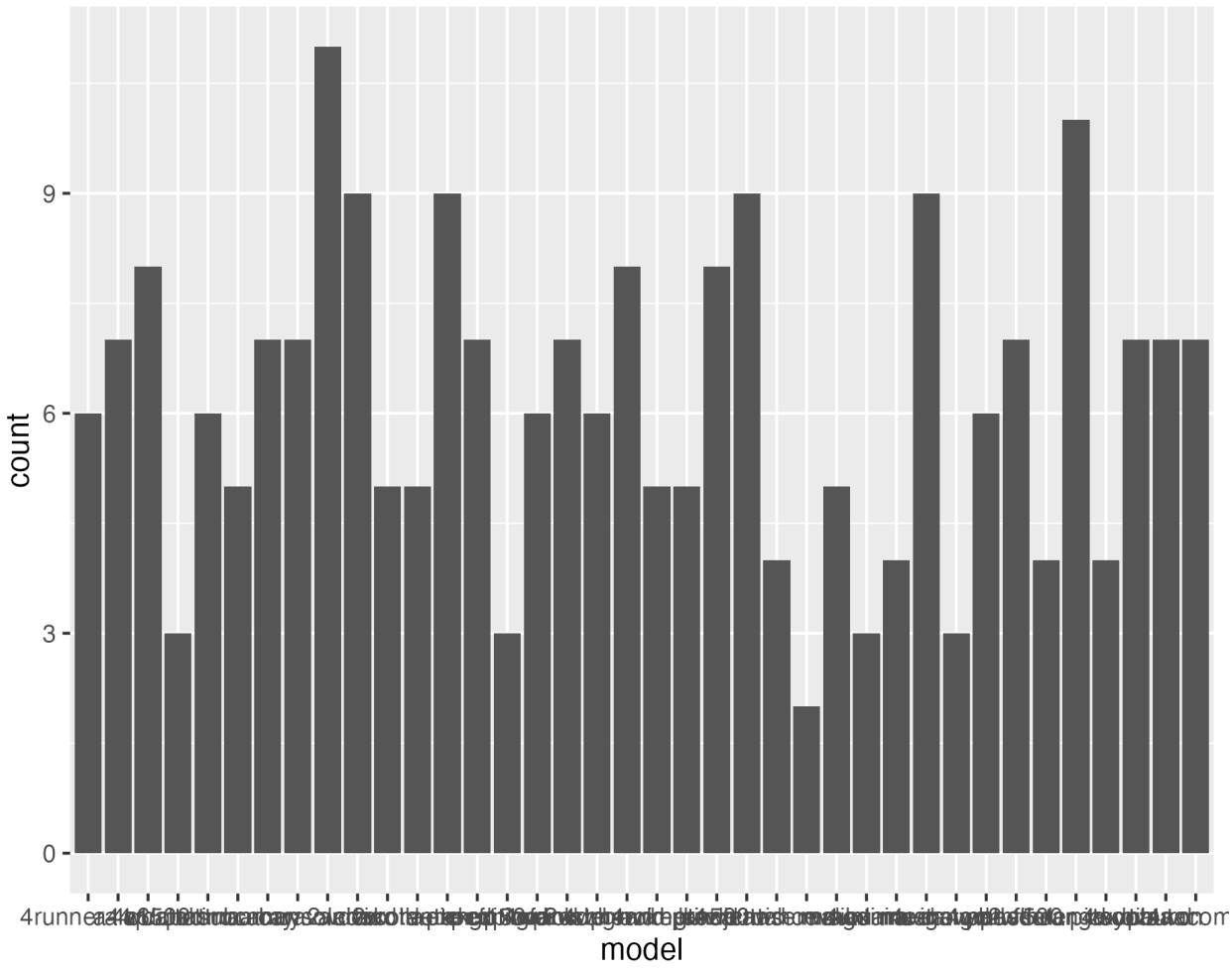
produces scatterplot of car models vs manufacturers

not very readable as both are categorical variables, thus it's just a messy grid with overlapping things

ggplot(mpg, aes(manufacturer)) + geom_bar()

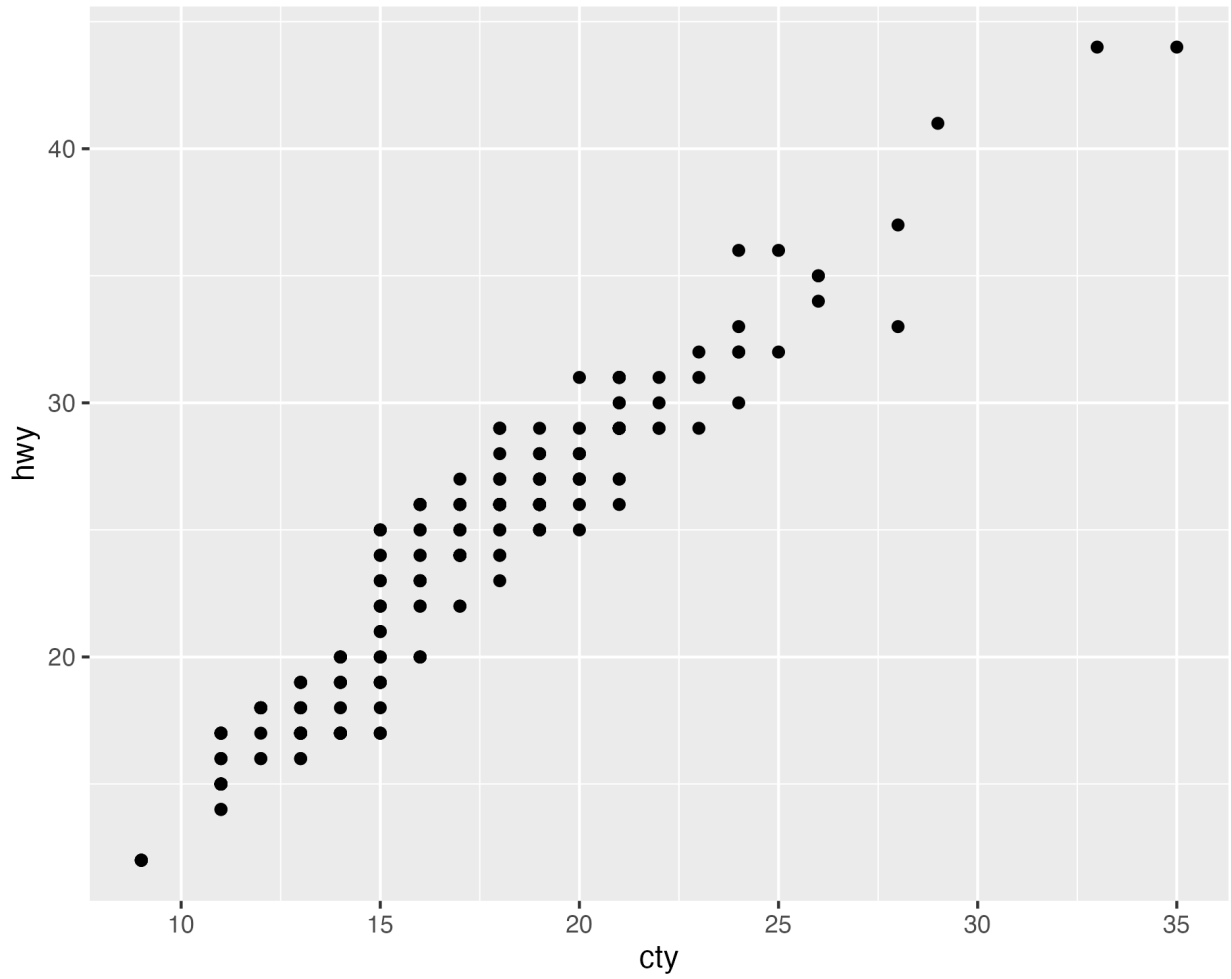



```
ggplot(mpg, aes(model)) + geom_bar()
```



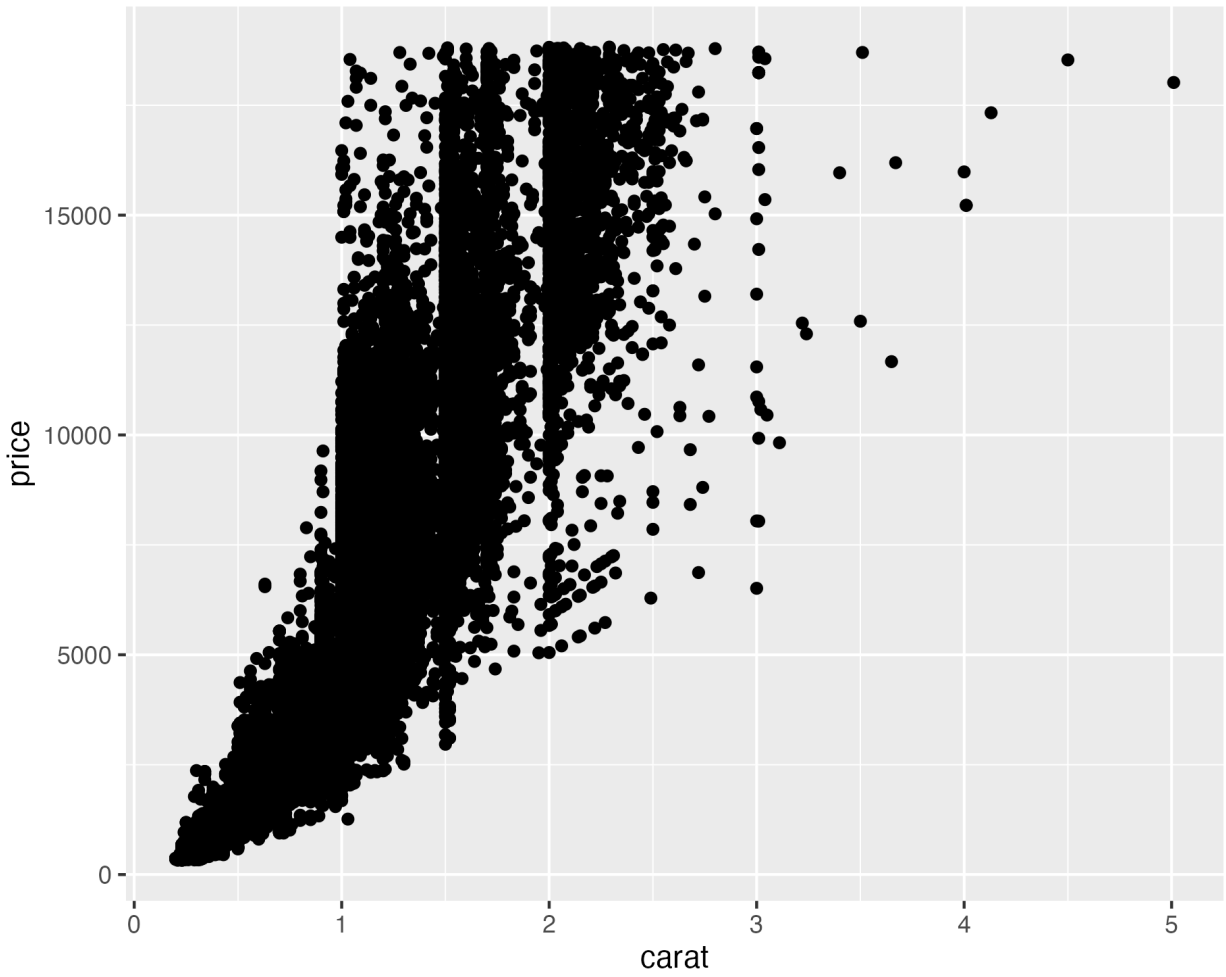
these might be more helpful, to visualize the count (a continuous variable) vs model/manufacturer.

ggplot(mpg, aes(cty, hwy)) + geom_point()



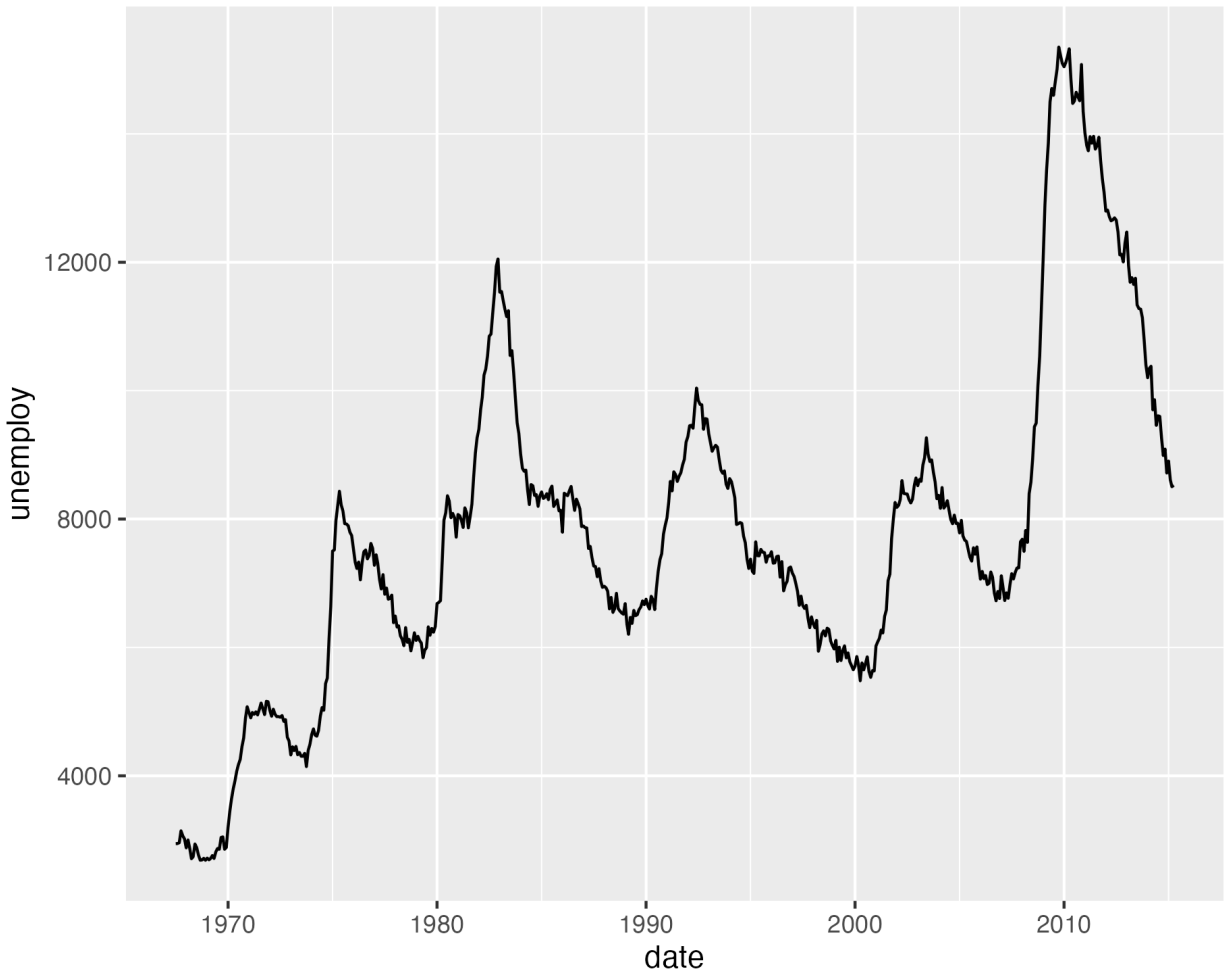
scatterplot detailing relationship between mpg for city and highway driving

ggplot(diamonds, aes(carat, price)) + geom_point()



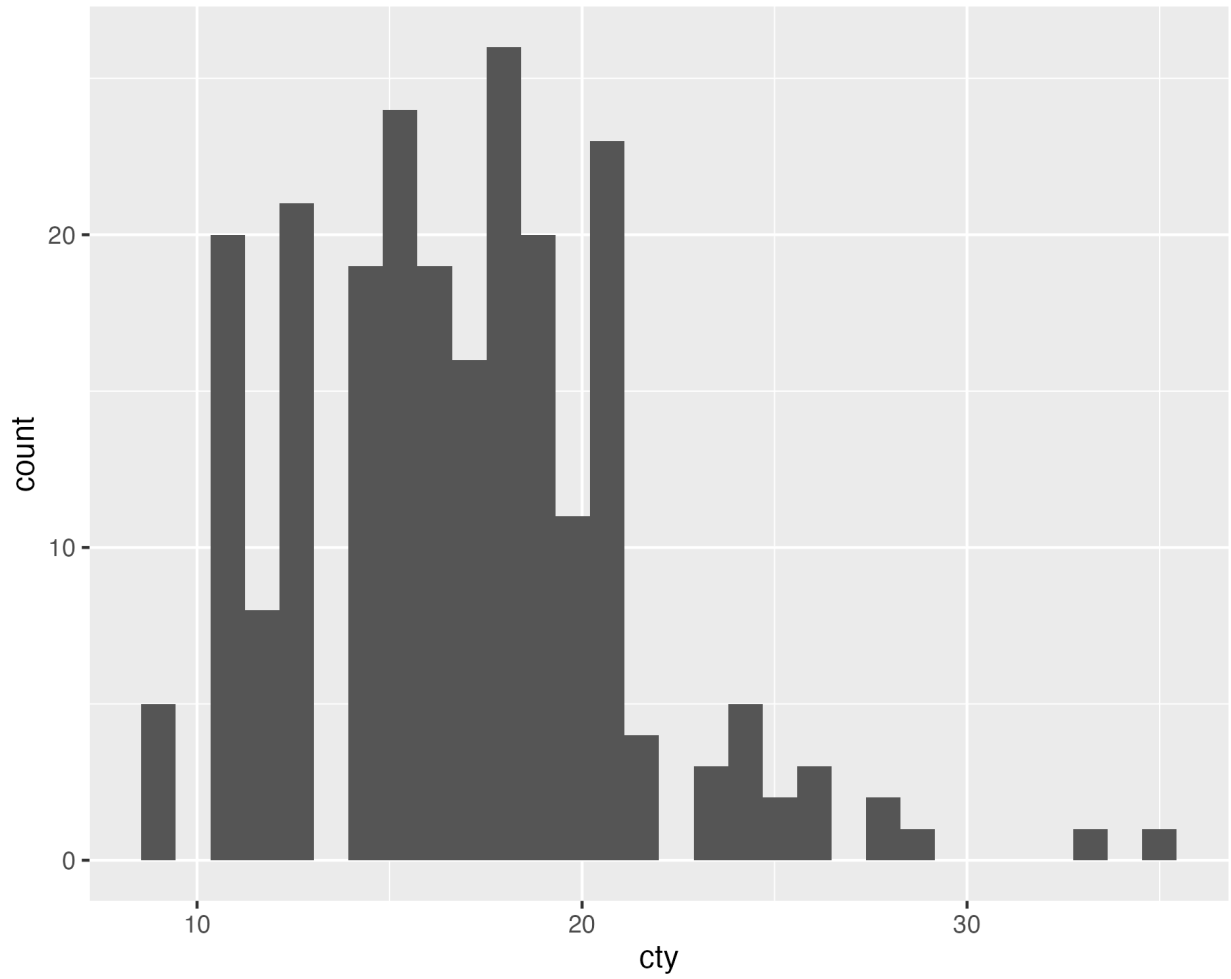
scatterplot detailing relationship between how many carats in a diamond and its price

ggplot(economics, aes(date, unemploy)) + geom_line()



line graph detailing the unemployment rate for the past 50 or so years

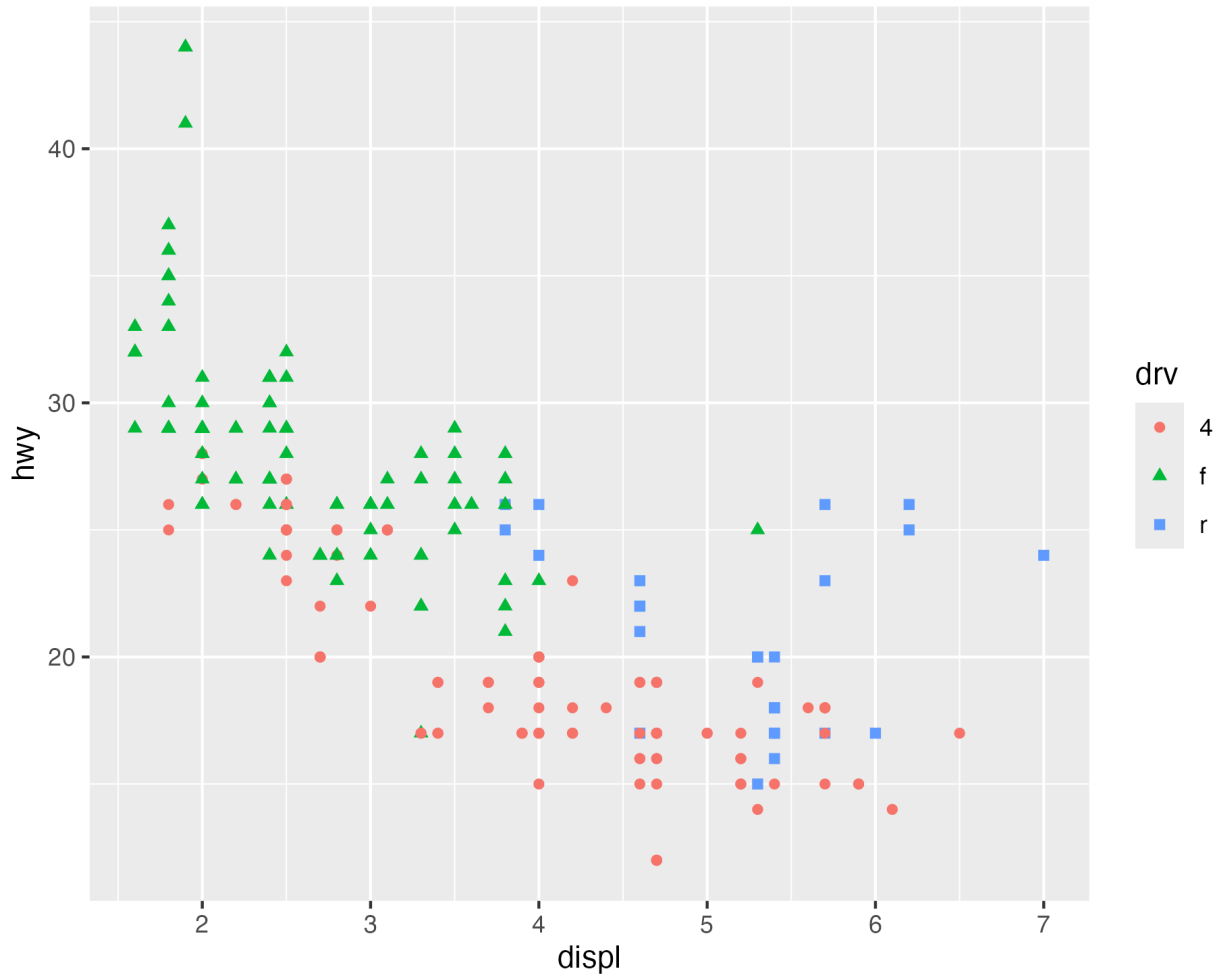
ggplot(mpg, aes(cty)) + geom_histogram()



histogram detailing the number of cars in the data set which correspond to specific city mpg values

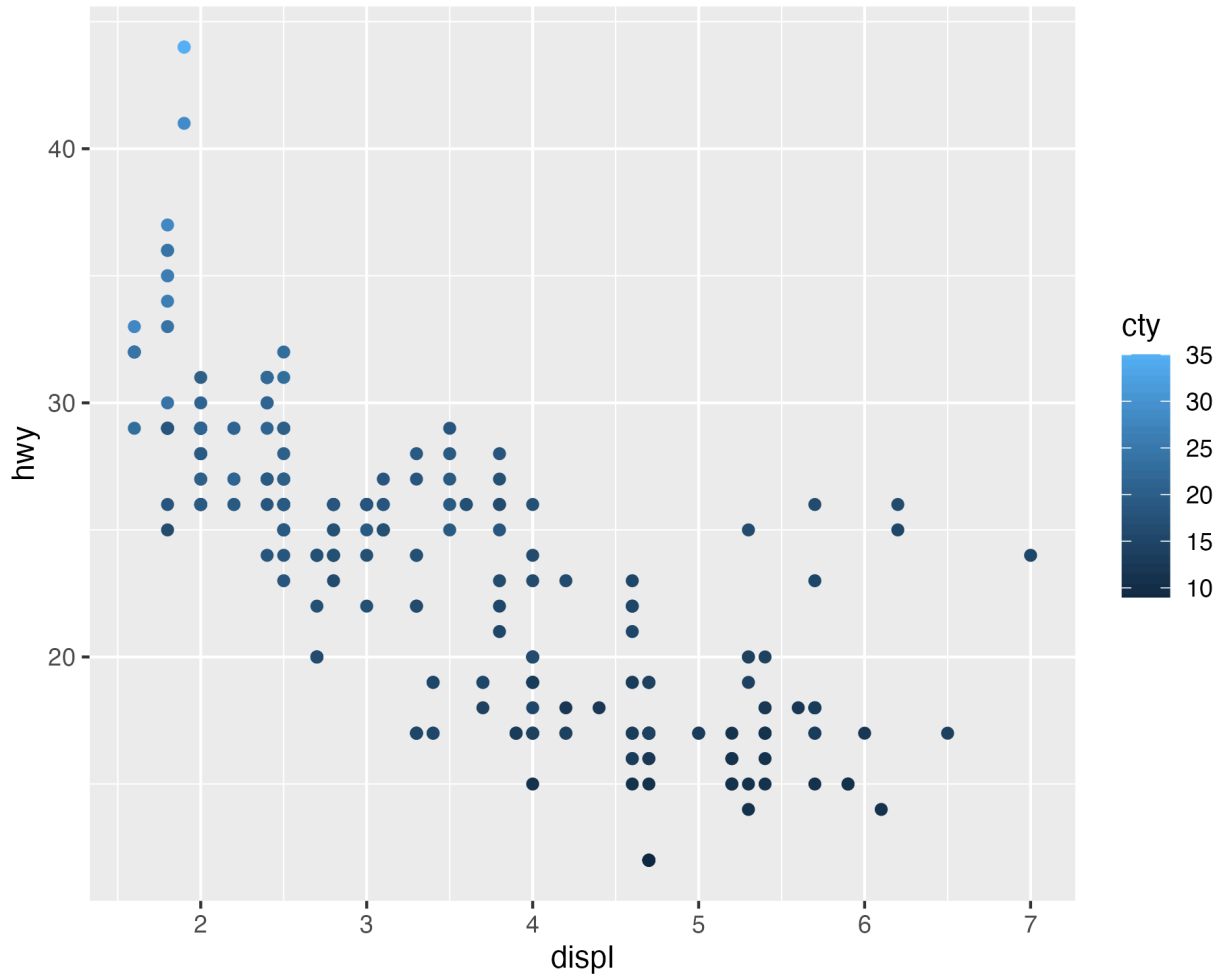
2.4.1 Exercises

ggplot(mpg, aes(displ, hwy, shape = drv, color = drv)) + geom_point()

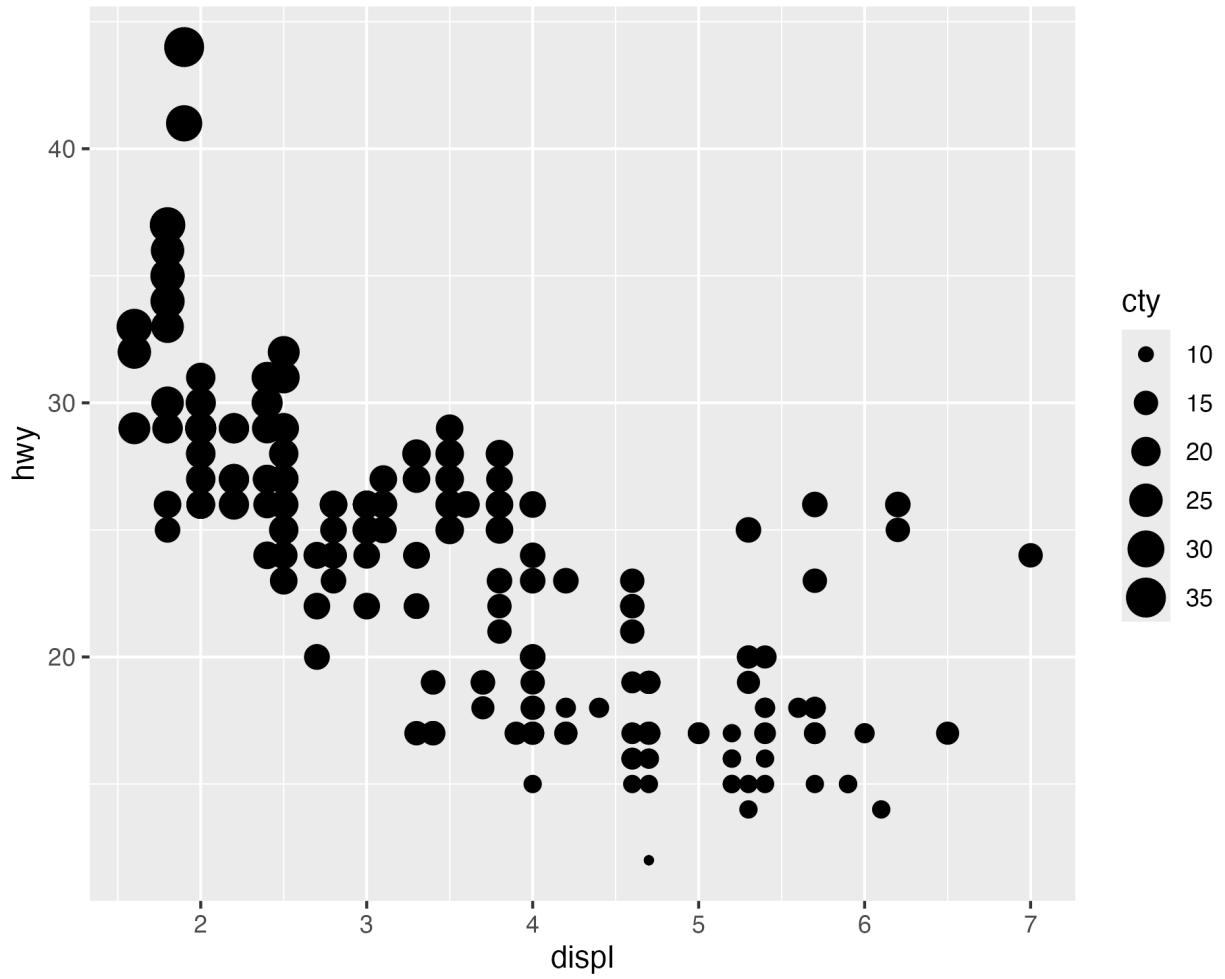


you can add multiple color, shape, size, aesthetics!

ggplot(mpg, aes(displ, hwy, color = cty)) + geom_point()



ggplot(mpg, aes(displ, hwy, size = cty)) + geom_point()

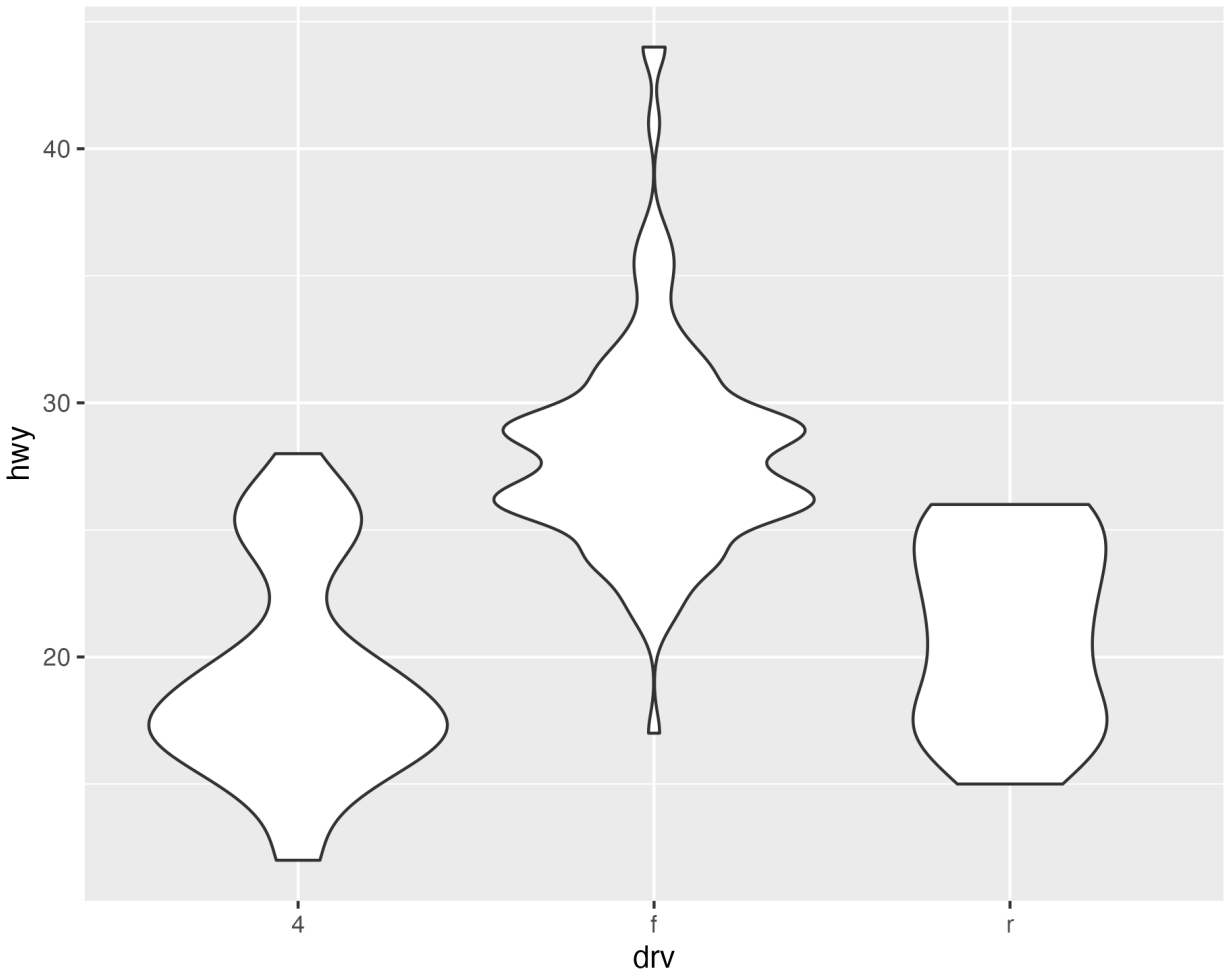


continuous variables can be mapped to color and size aesthetics

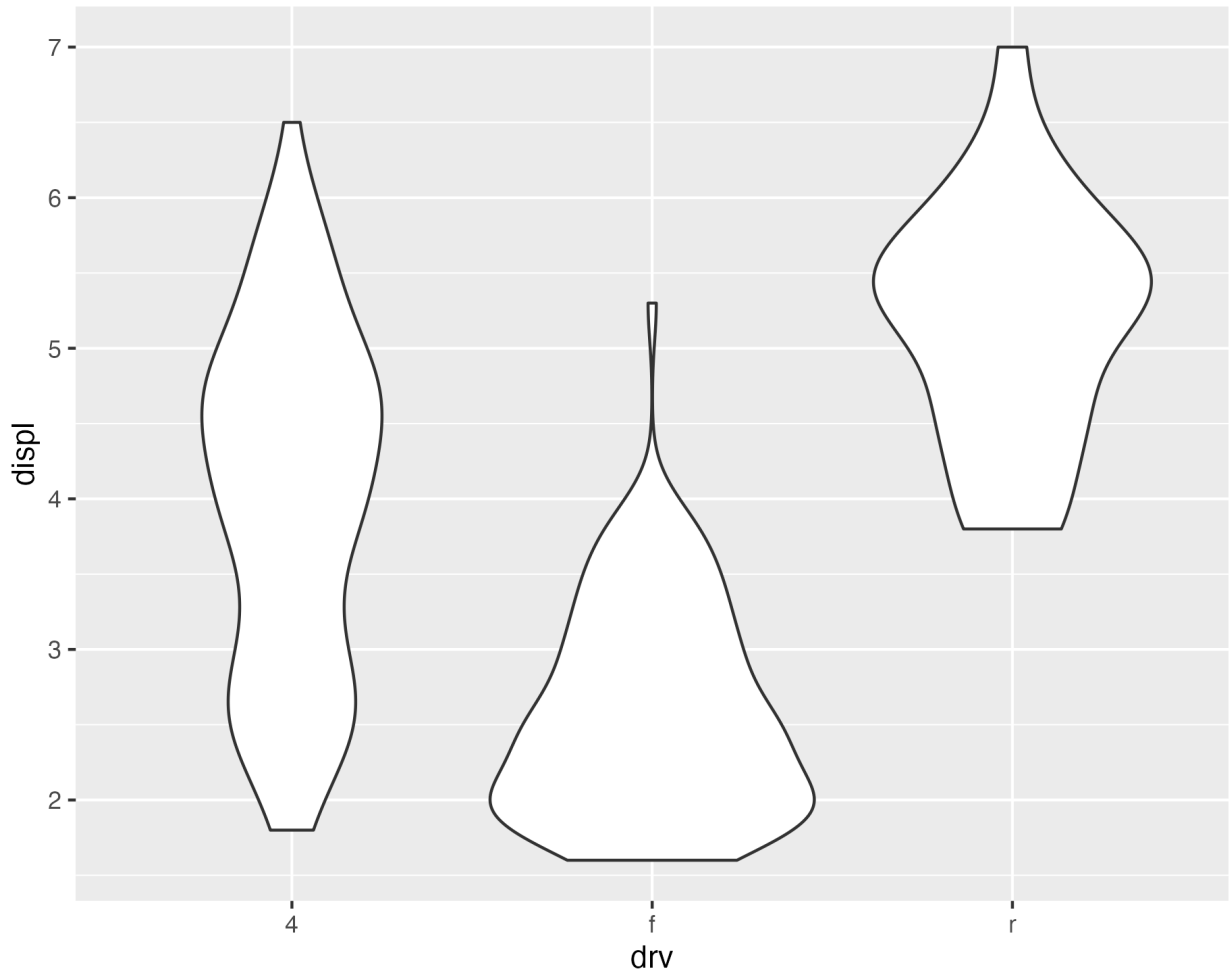
ggplot(mpg, aes(displ, hwy, shape = cty)) + geom_point()

however, "a continuous variable cannot be mapped to the shape aesthetic"

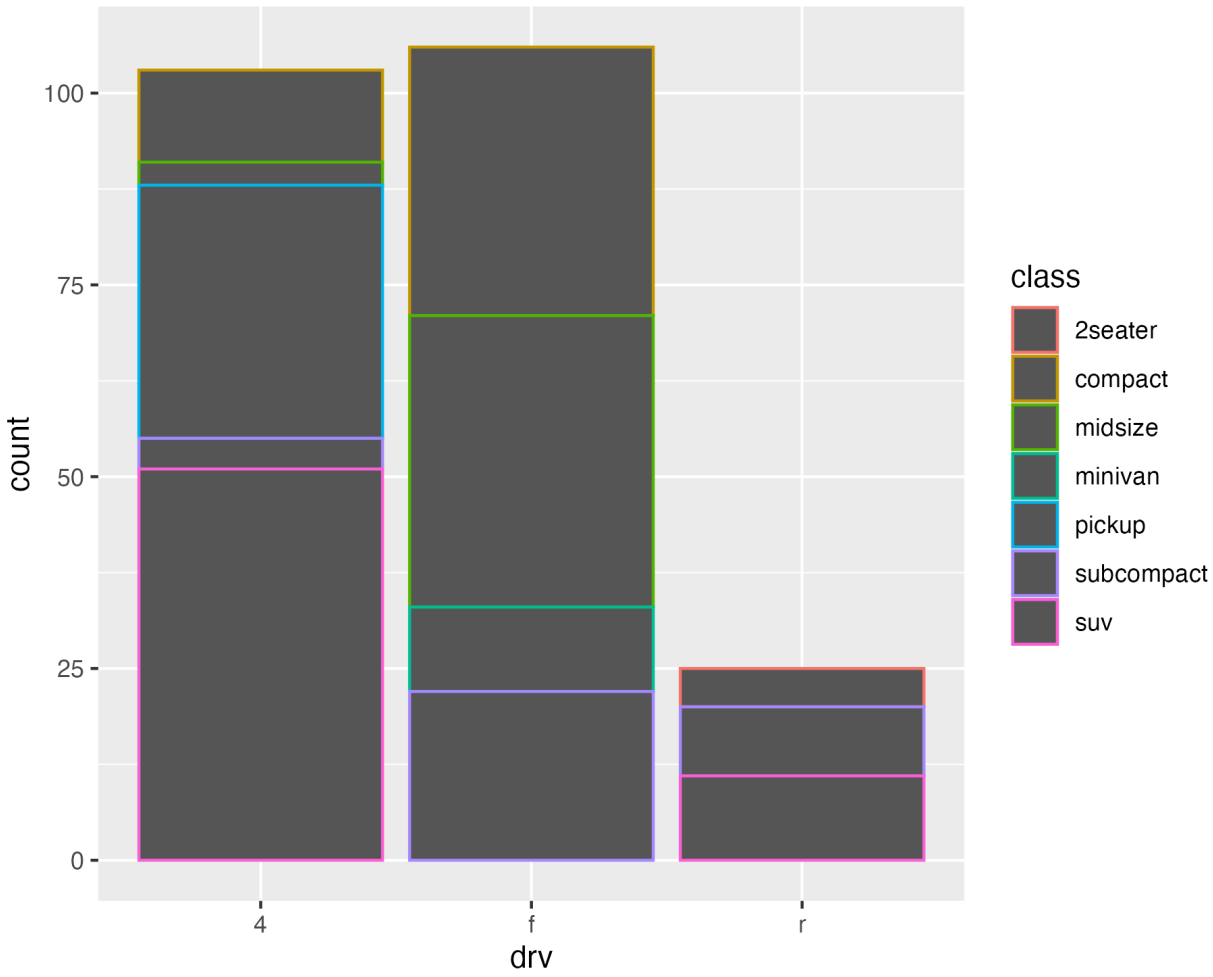
ggplot(mpg, aes(drv, hwy)) + geom_violin()




```
# f has best fuel economy, then r, then 4
ggplot(mpg, aes(drv, displ)) + geom_violin()
```



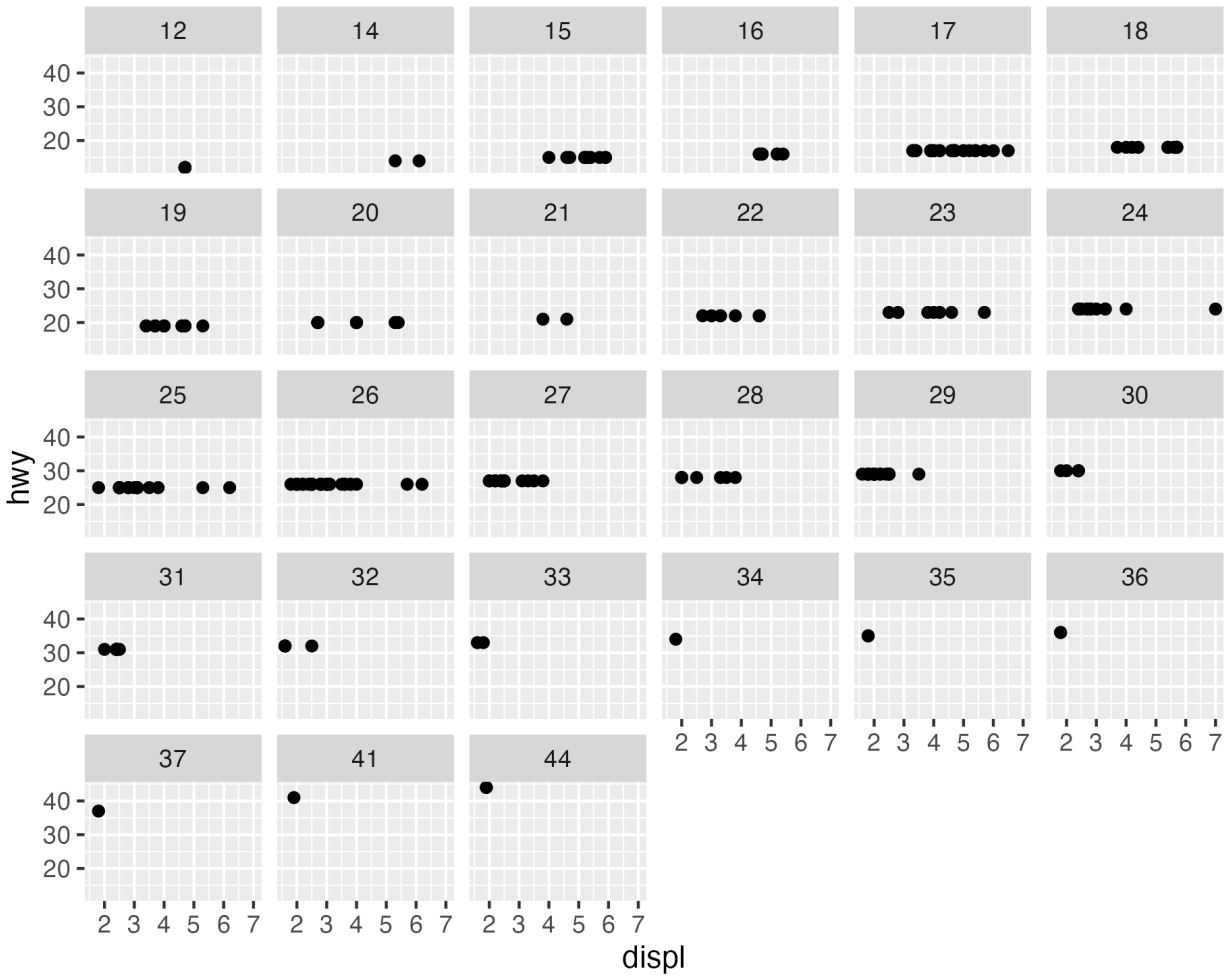

```
# r is associated with large engines, then 4 then f
ggplot(mpg, aes(drv, color = class)) + geom_bar()
```



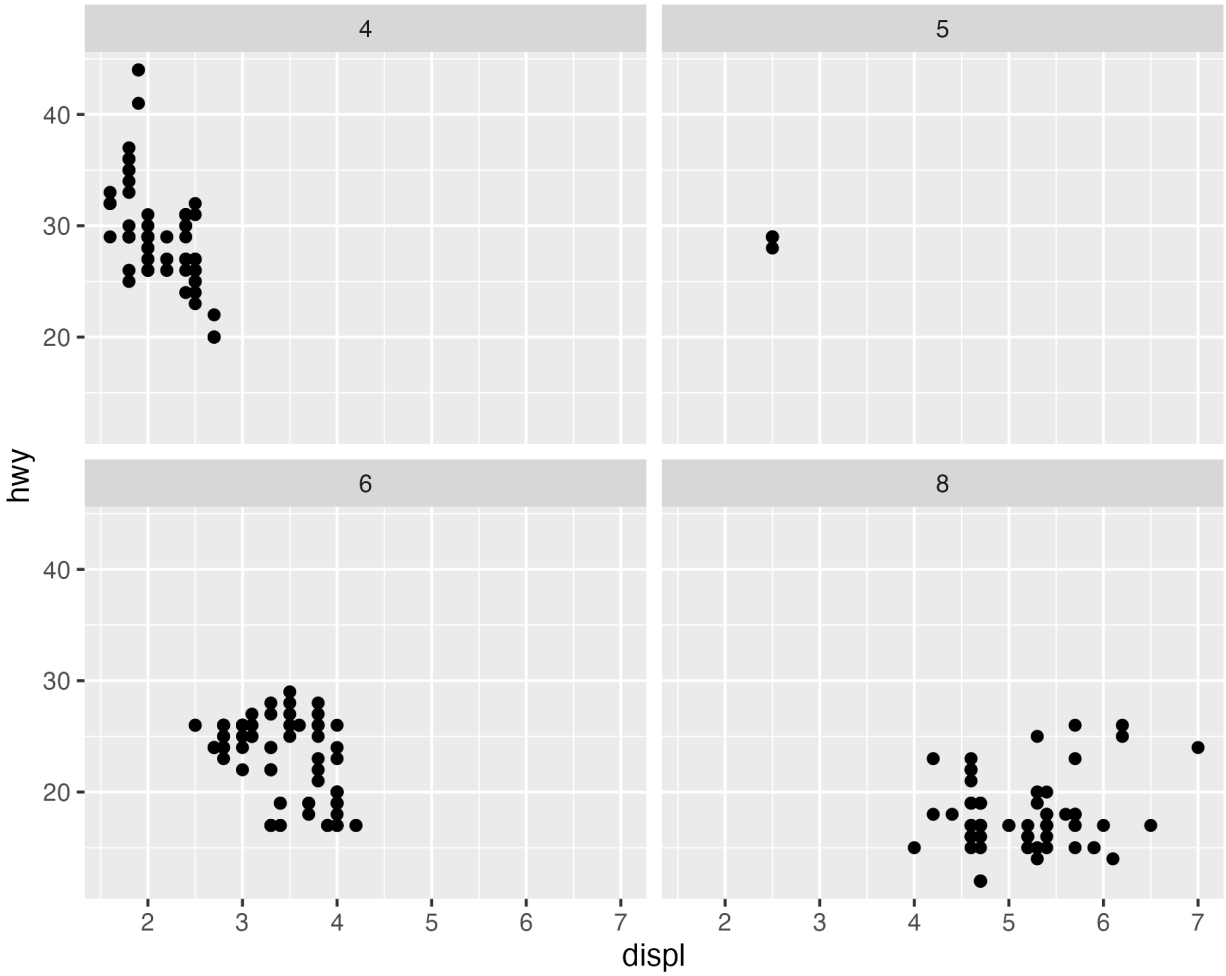
4 is mostly suvs, f is mostly compact/midsize, r is mostly subcompact/suvs

2.5.1 Exercises

```
ggplot(mpg, aes(displ, hwy)) +
  geom_point() +
  facet_wrap(~hwy)
```

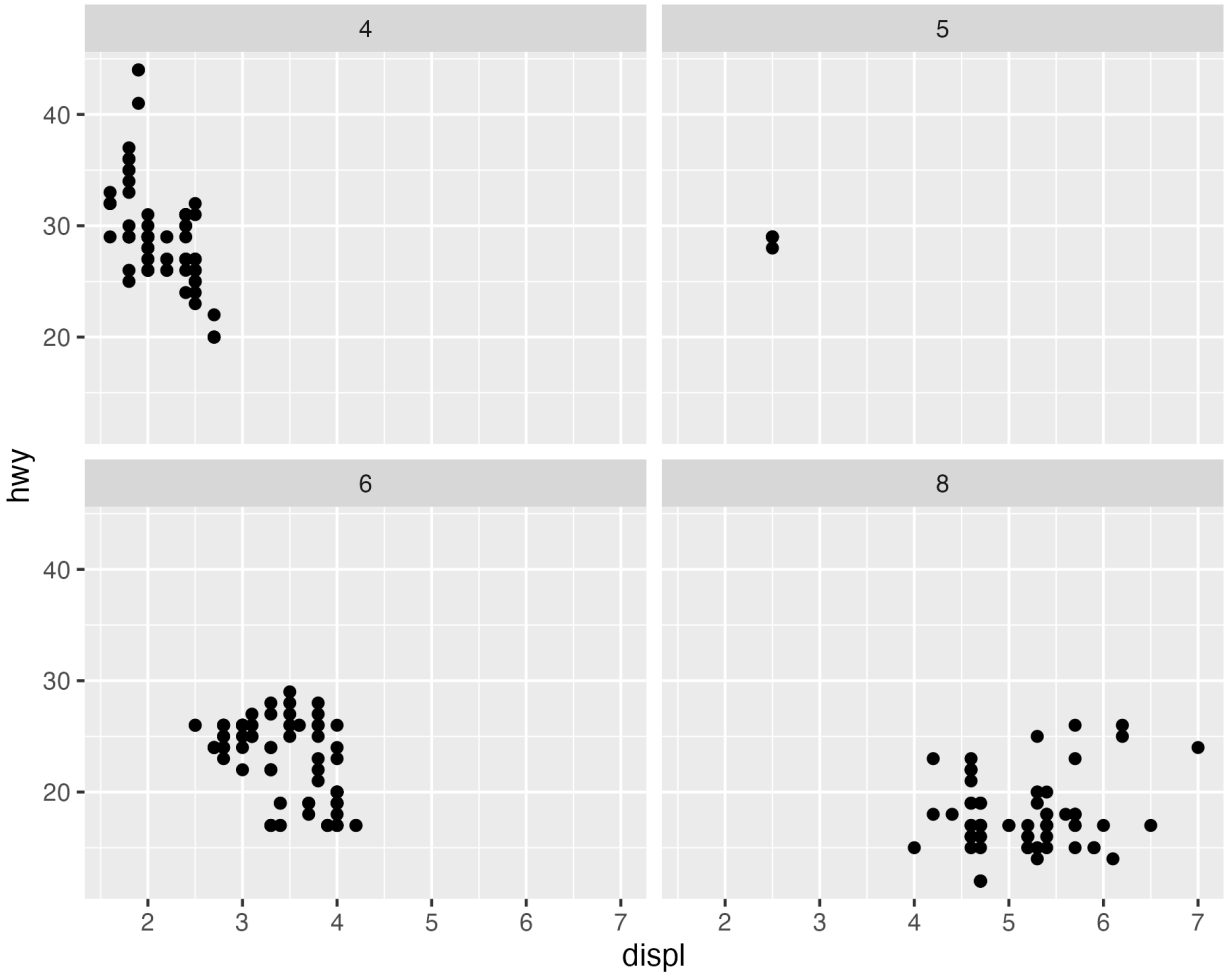



```
ggplot(mpg, aes(displ, hwy)) +
  geom_point() +
  facet_wrap(~cyl)
```



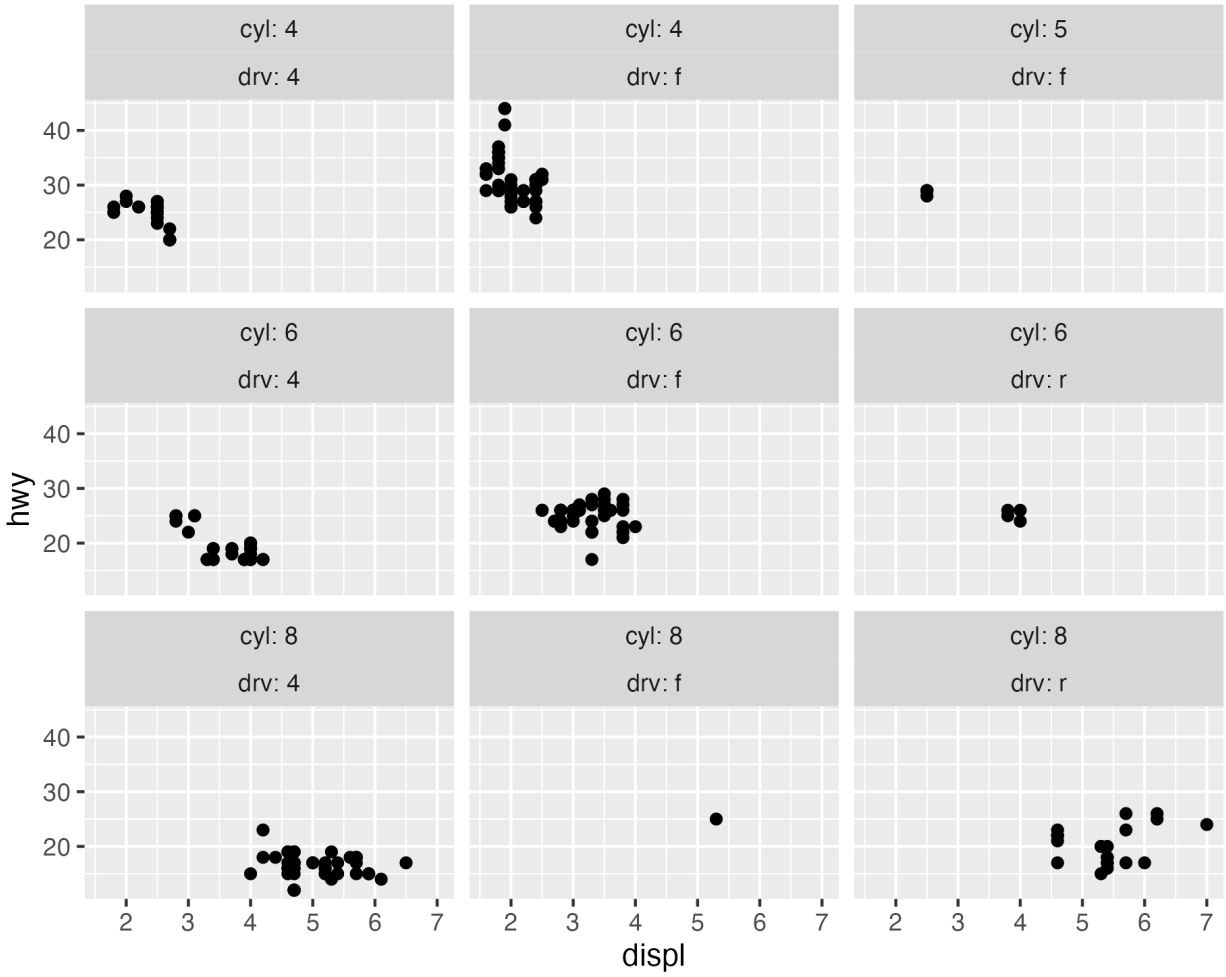

```
# when faceting with continuous variables, there are quite a bit more graphs that it displays

ggplot(mpg, aes(displ, hwy)) +
  geom_point() +
  facet_wrap(~cyl)
```



faceting by number of cylinders, shows clusters in different spots in the graphs between displ vs hwy
this probably means that the number of cylinders contributes quite a bit to the differences?

```
ggplot(mpg, aes(displ, hwy)) +
  geom_point() +
  facet_wrap(vars(cyl, drv), nrow=4, labeller = "label_both")
```

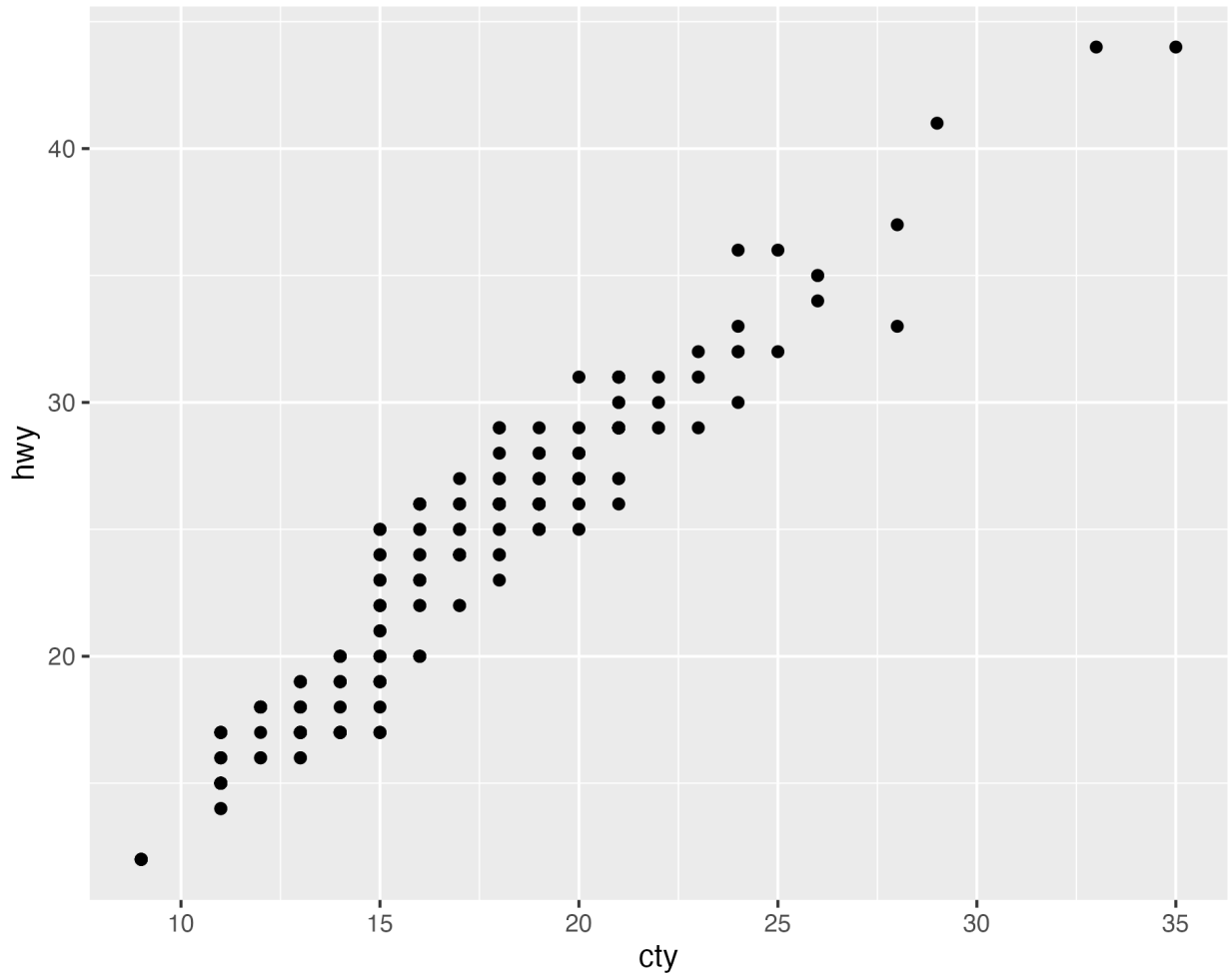


from documentation: can set number of rows + facet by multiple variables + labels, etc...

scales: if scales should be fixed ("fixed") or free ("free") or free in one dimension ("free_x" or "free_y")
might use if you want to emphasize change in one dimension more than another.

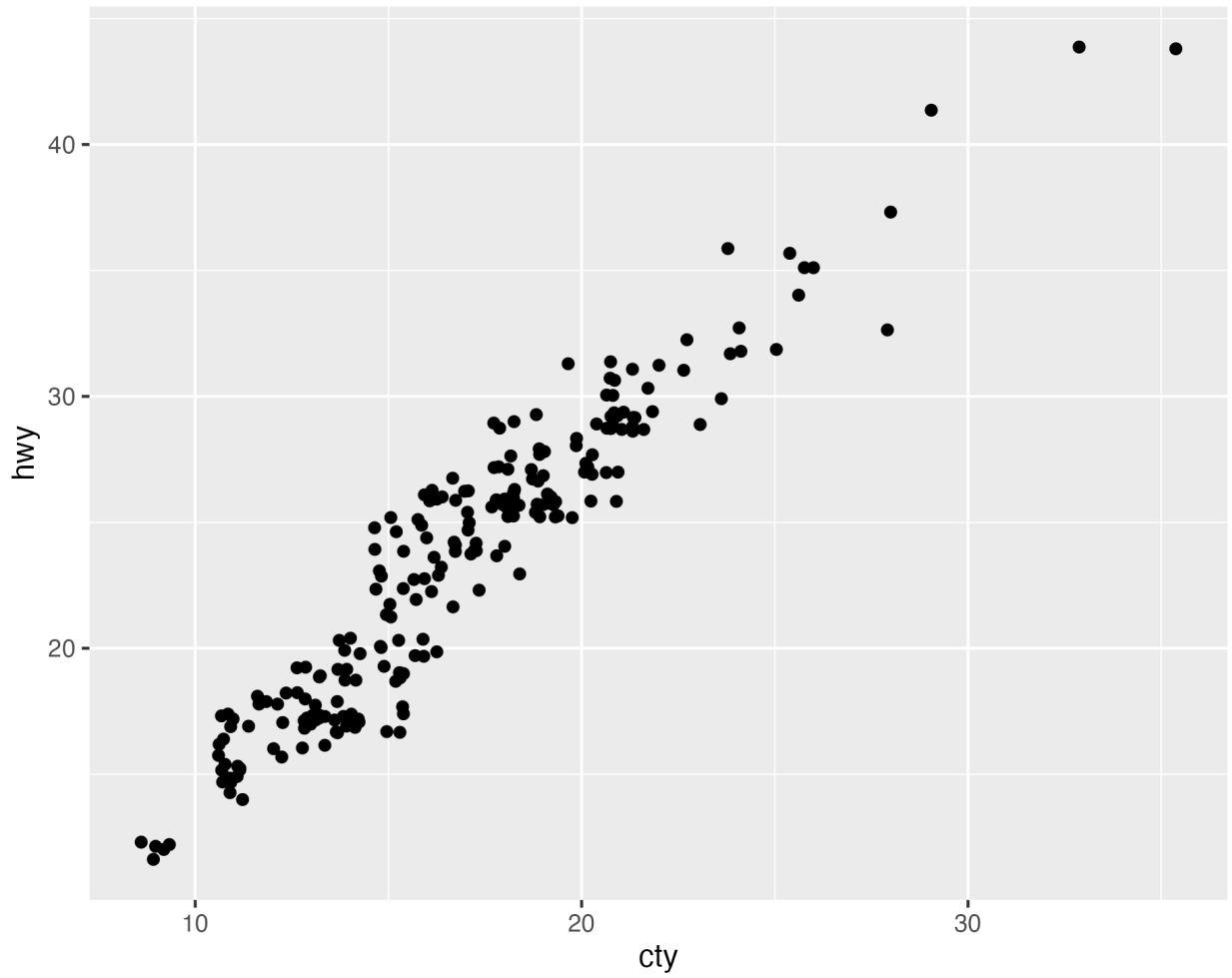
2.6.6 Exercises

ggplot(mpg, aes(cty, hwy)) + geom_point()

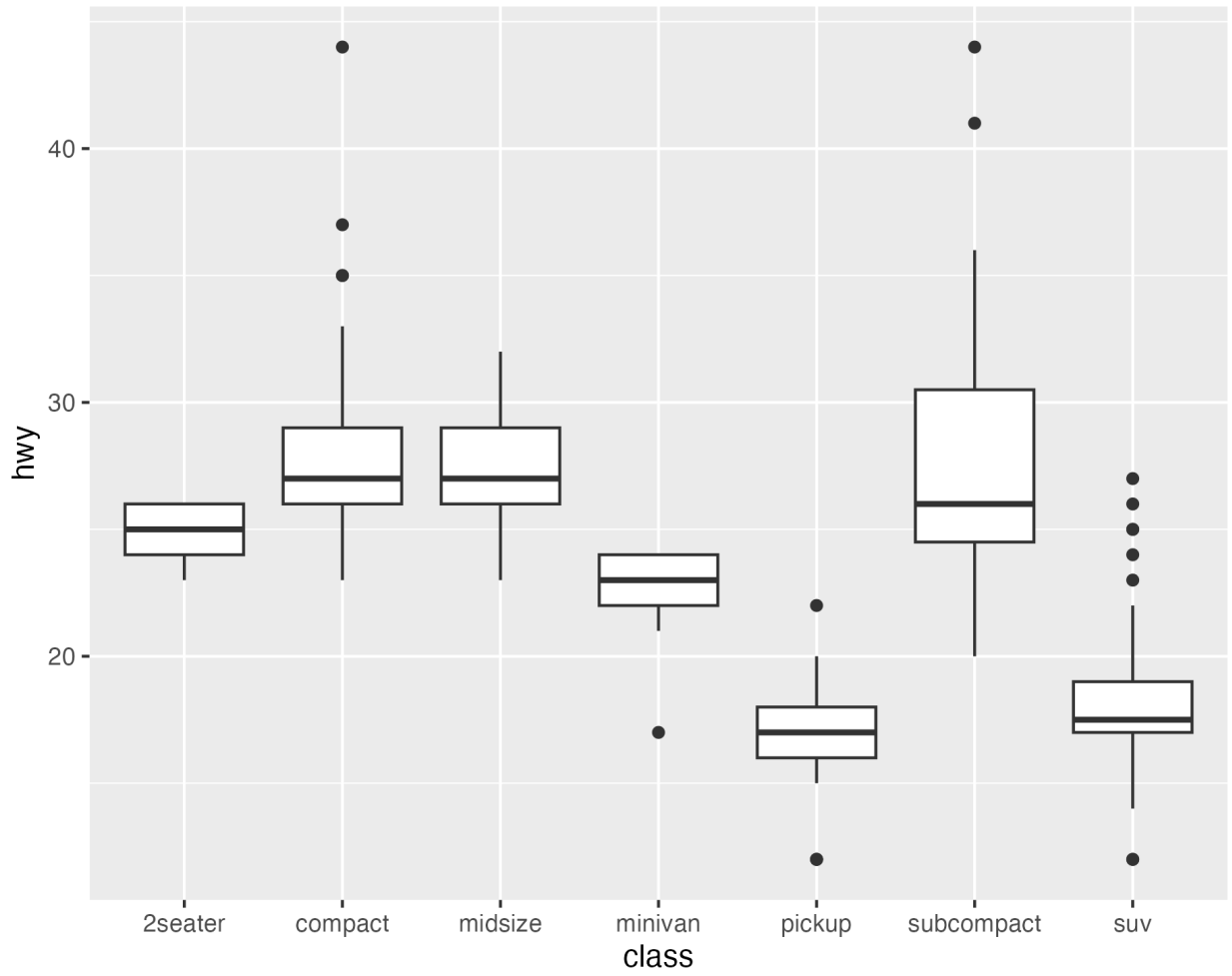


lots of overlap, using jitter would improve readability:

ggplot(mpg, aes(cty, hwy)) + geom_jitter()

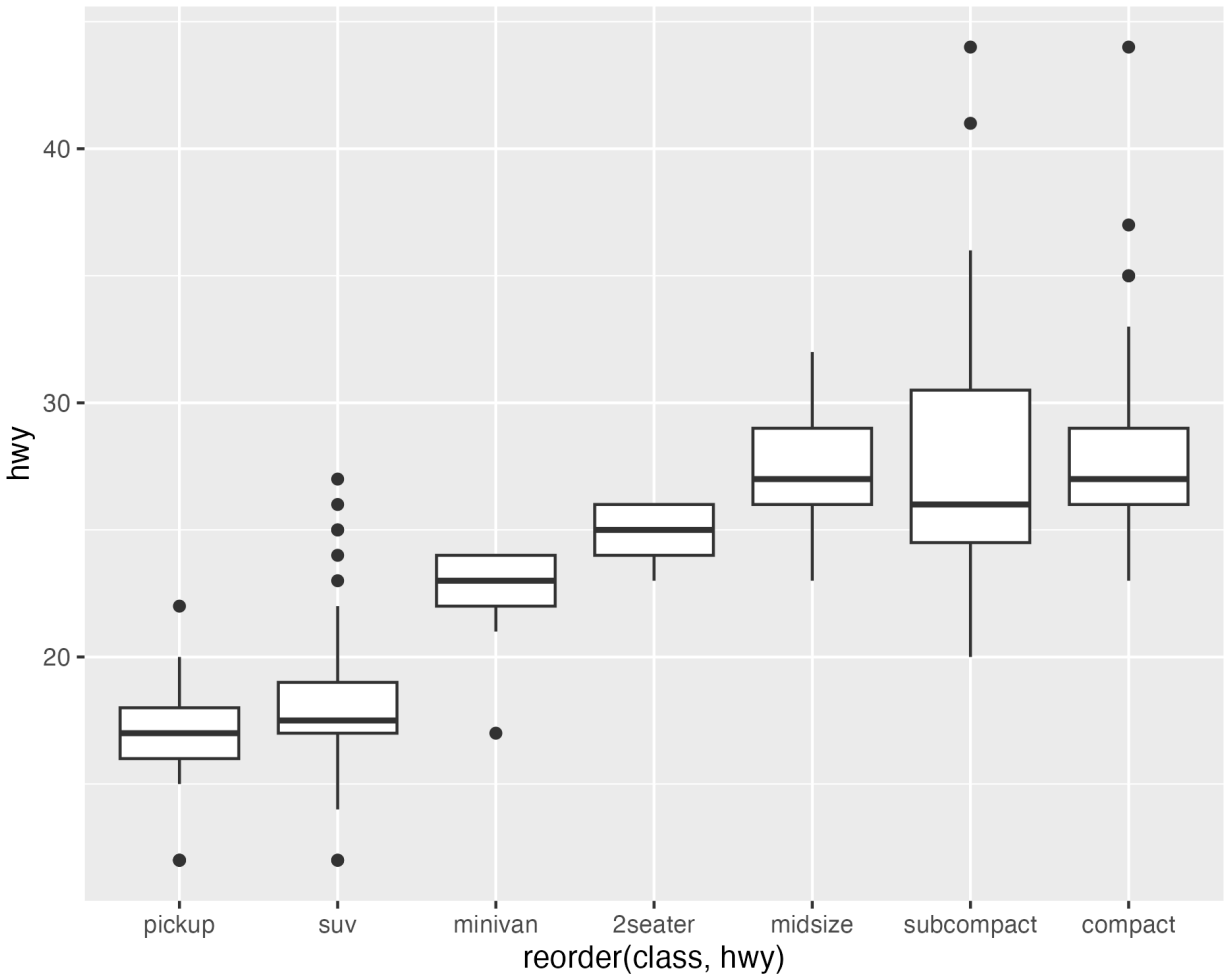


ggplot(mpg, aes(class, hwy)) + geom_boxplot()



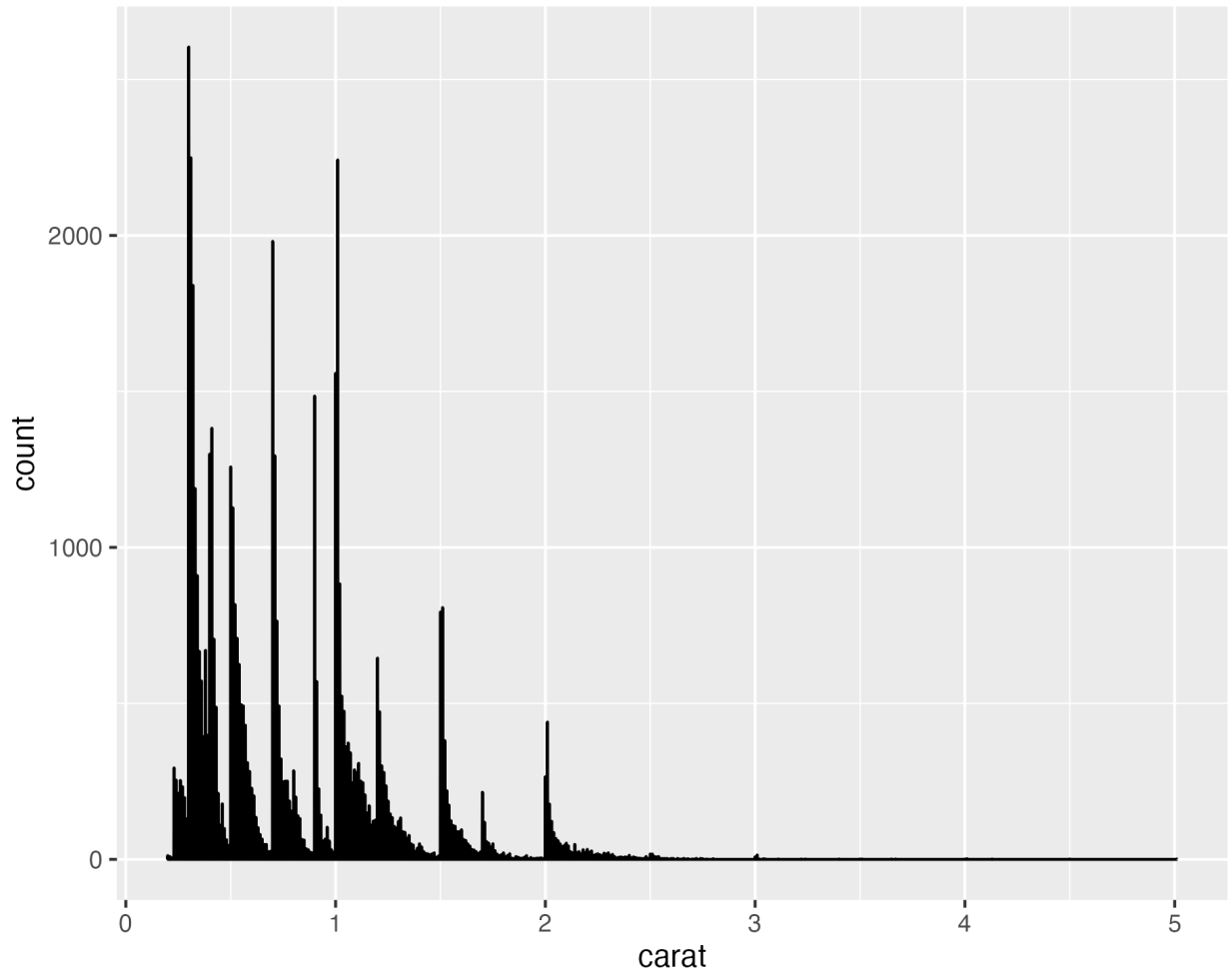
you could order them by median highway mpg

ggplot(mpg, aes(reorder(class, hwy), hwy)) + geom_boxplot()



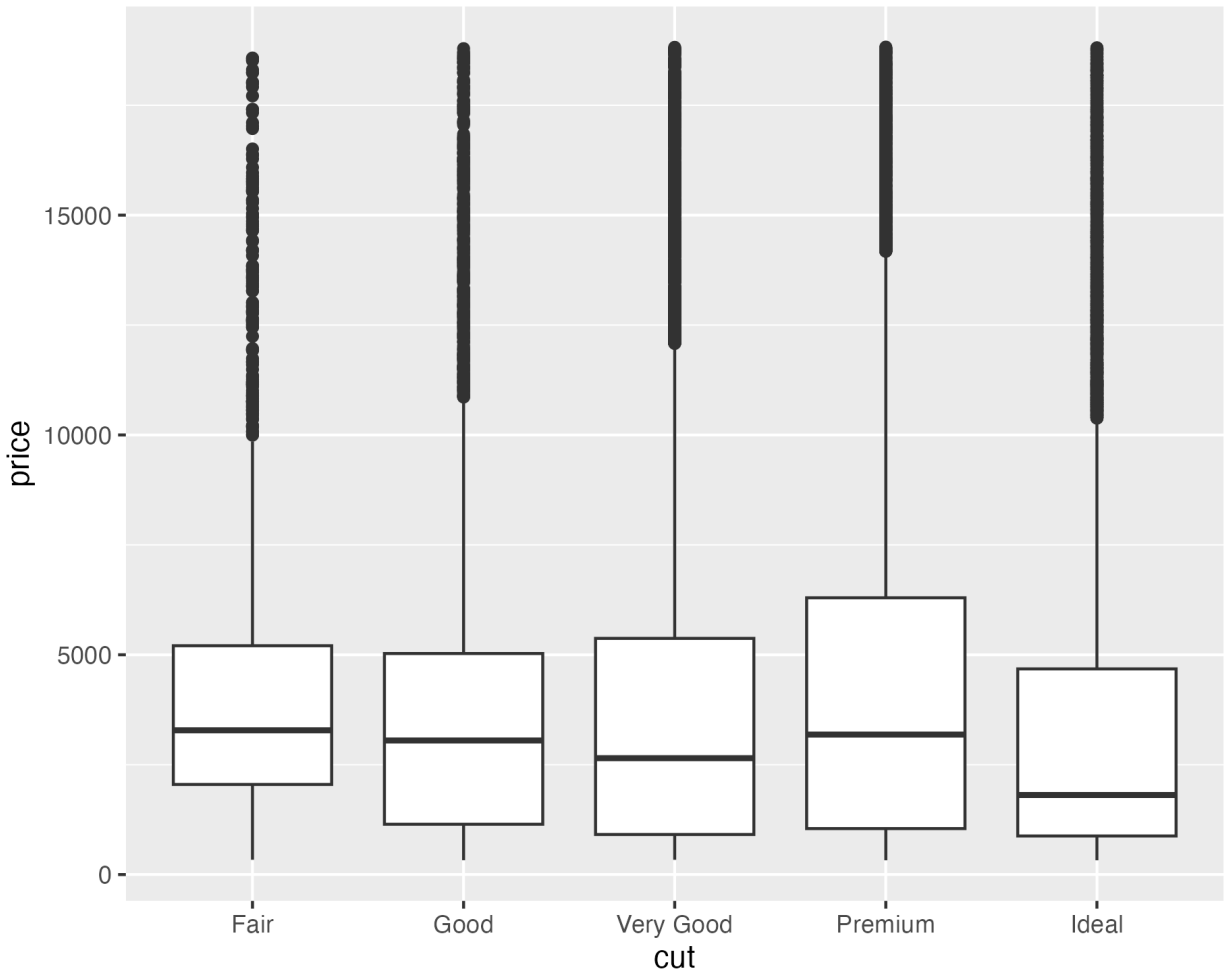
reorder takes the variable and reorders the levels based on a variable (default is median)

ggplot(diamonds, aes(carat)) + geom_freqpoly(binwidth = 0.001)



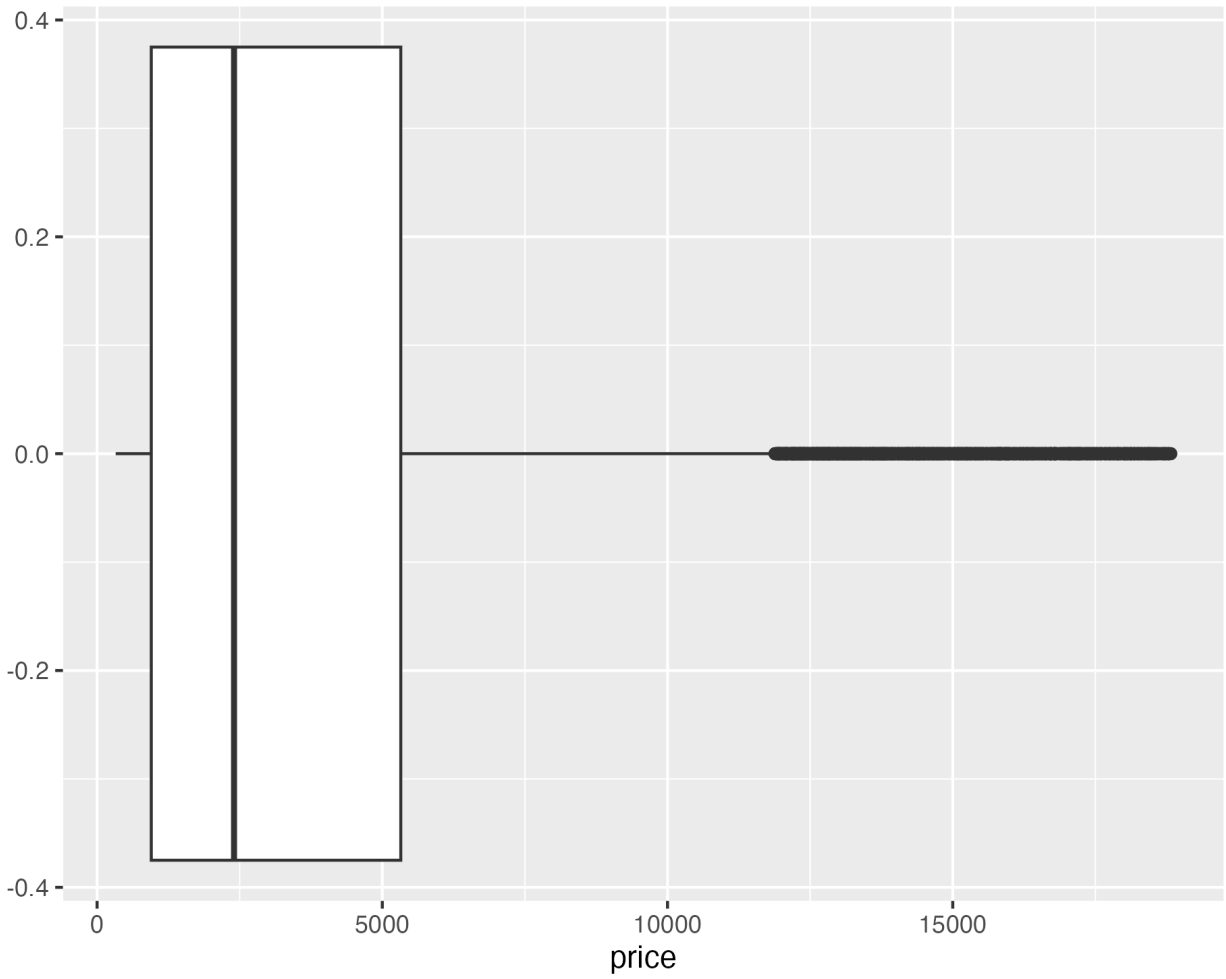
the smaller binwidths reveal more, showing that there are the most slightly right of specific carat marks
for example, 1 carat, 1.5 carat, 2 carat, etc...

ggplot(diamonds, aes(cut, price)) + geom_boxplot()



looks like fair cut diamonds have the highest median price, where ideal is the lowest

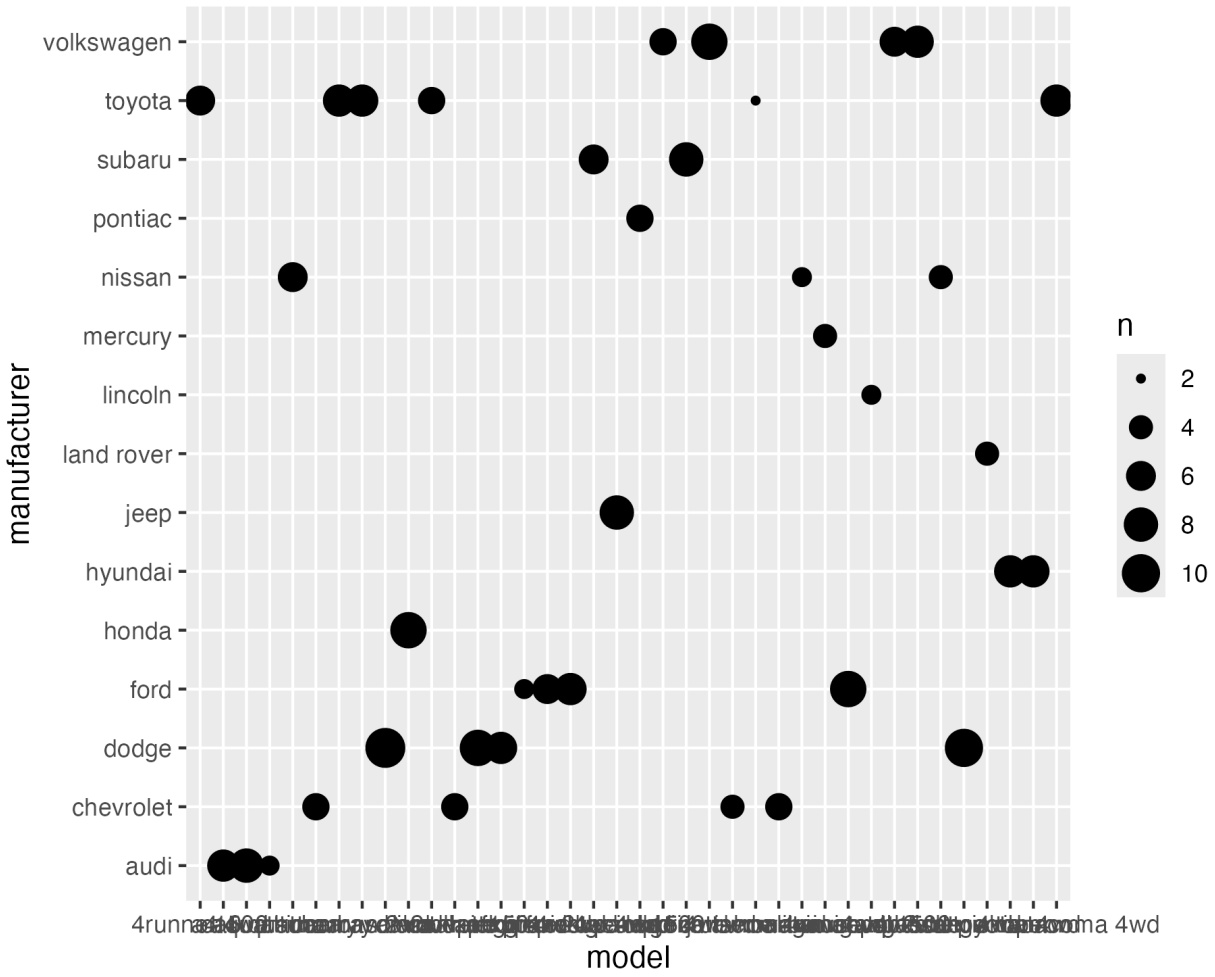
ggplot(diamonds, aes(price)) + geom_boxplot()



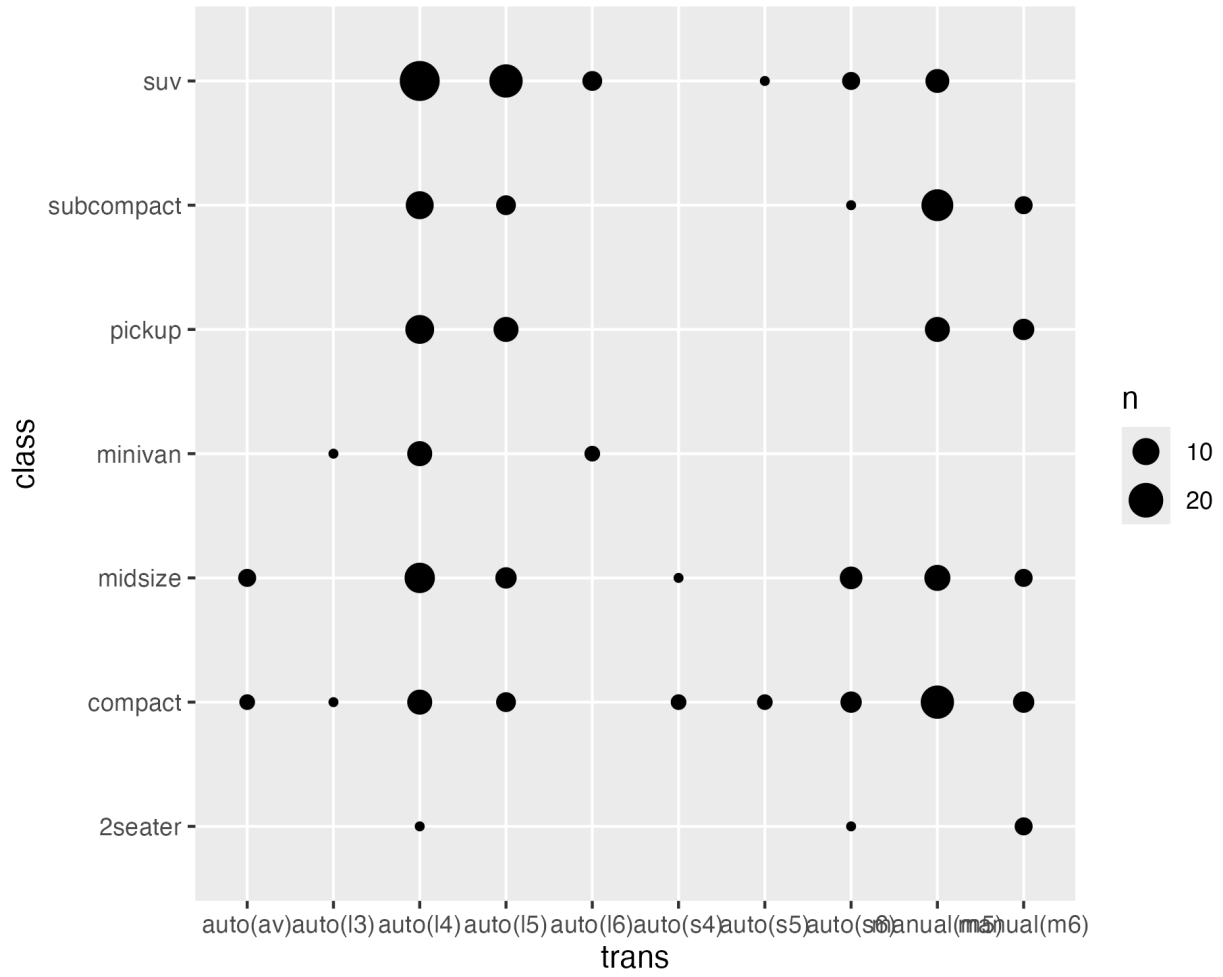
geom_violin --> looks the best, but not very specific
geom_freqpoly --> easy to see the overall trend (where the peaks are), but, can get kinda cramped
geom_histogram --> pretty accurate, but hard to see small differences

weight in geom_bar() makes it so that you can make bar charts where the height is a total value as opposed to just frequency

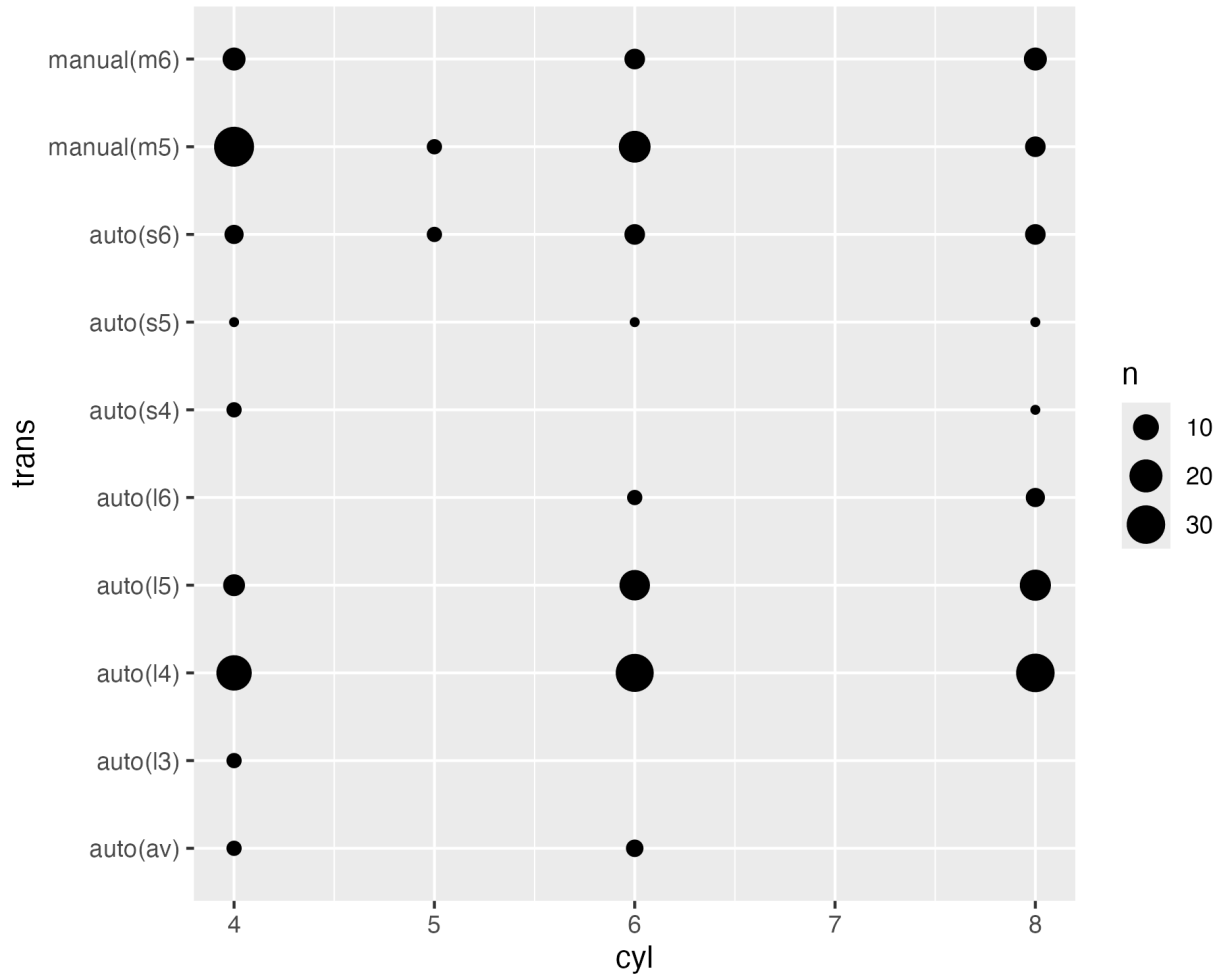
ggplot(mpg, aes(model, manufacturer)) + geom_count() # can use count (different sizes at the intersections)



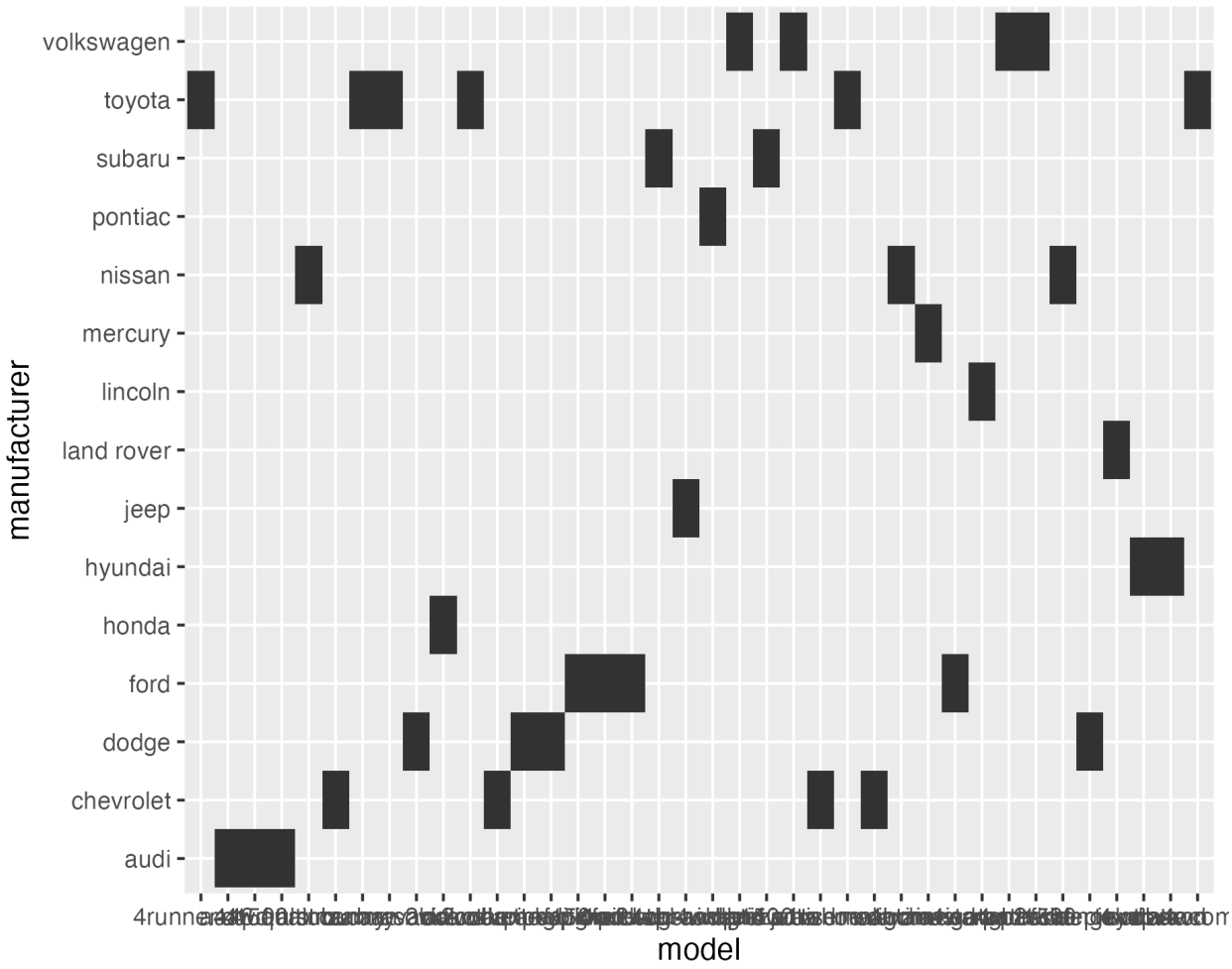

```
ggplot(mpg, aes(trans, class)) + geom_count()
```



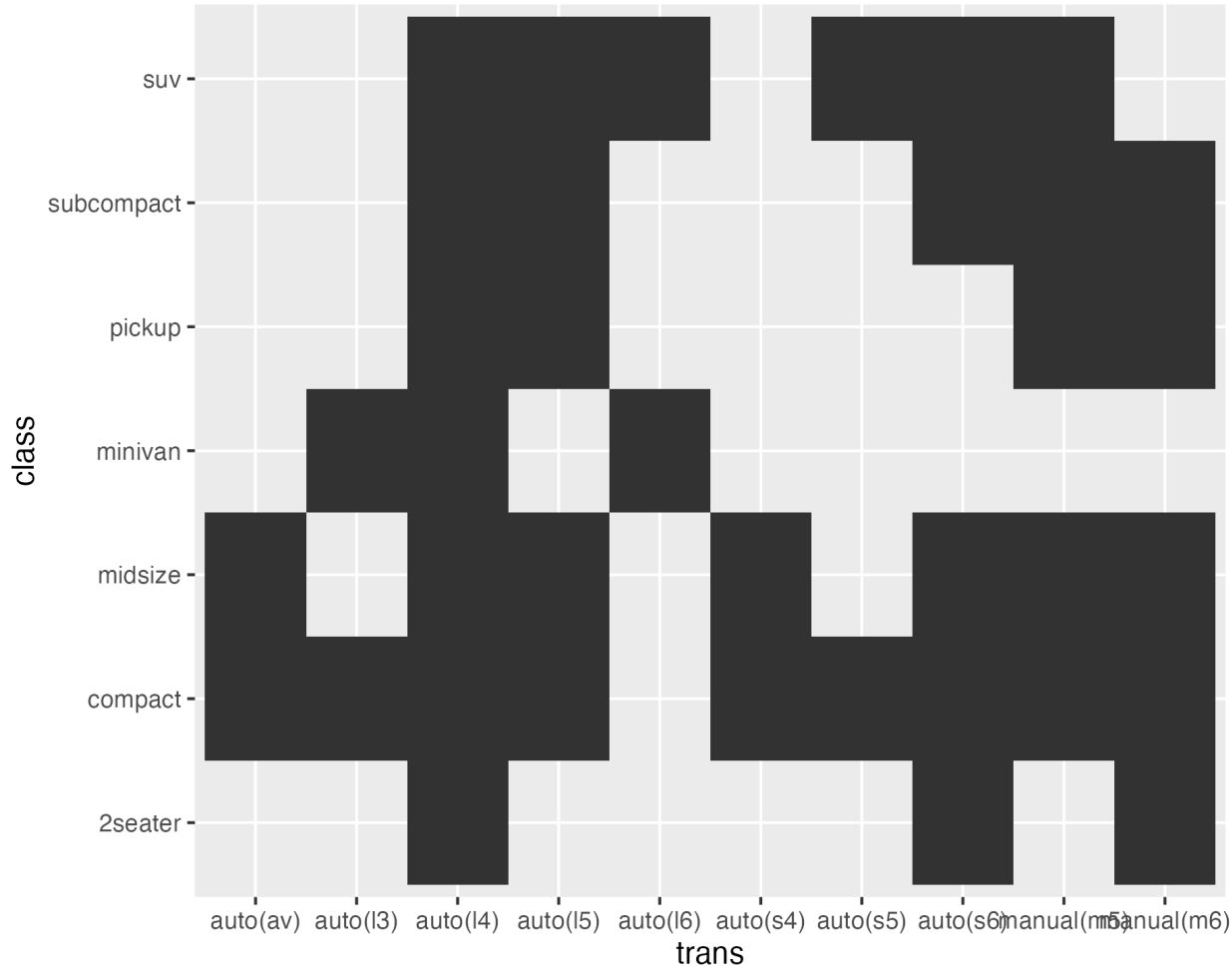
ggplot(mpg, aes(cyl, trans)) + geom_count()



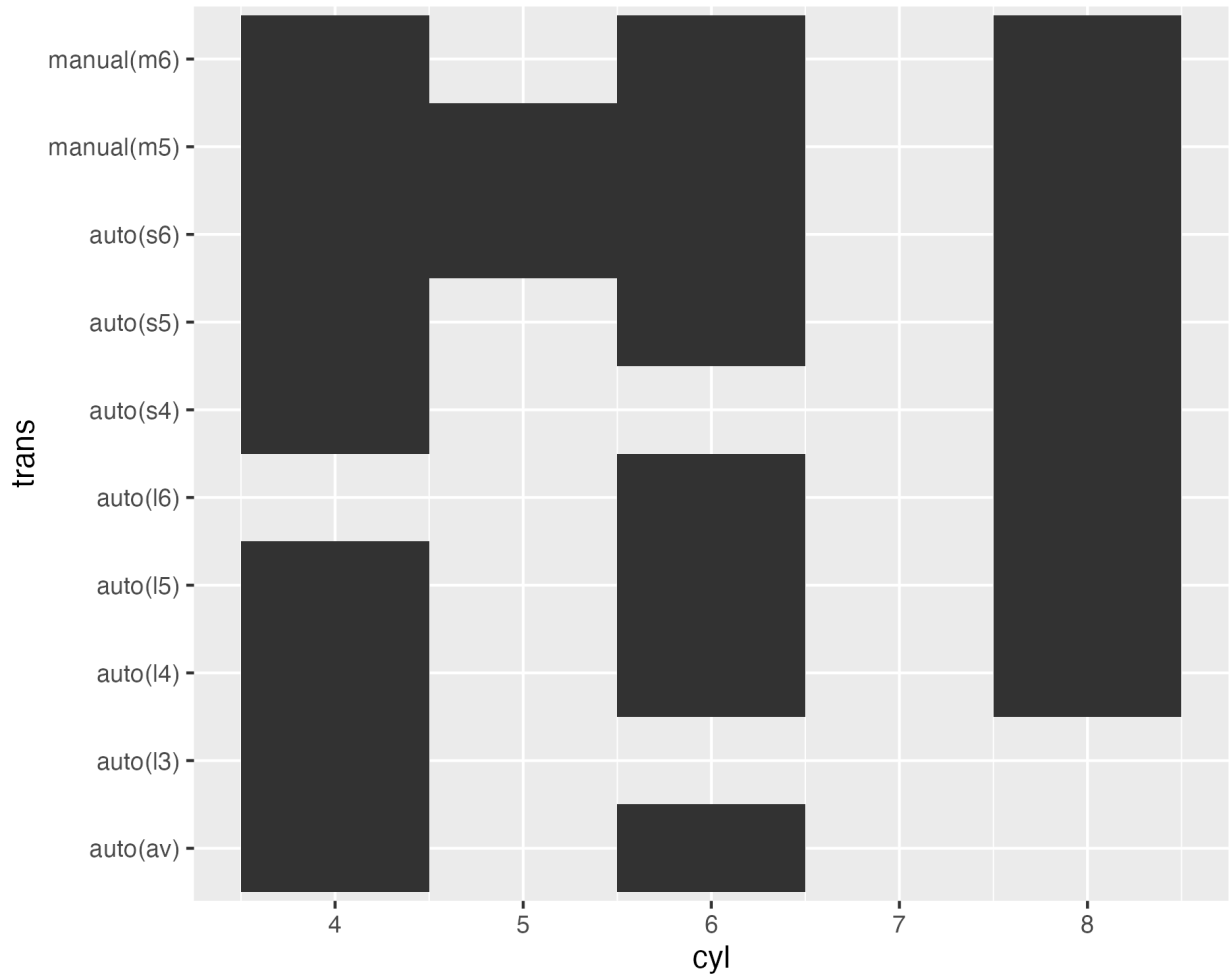
ggplot(mpg, aes(model, manufacturer)) + geom_tile() # can use this heatmap type thing



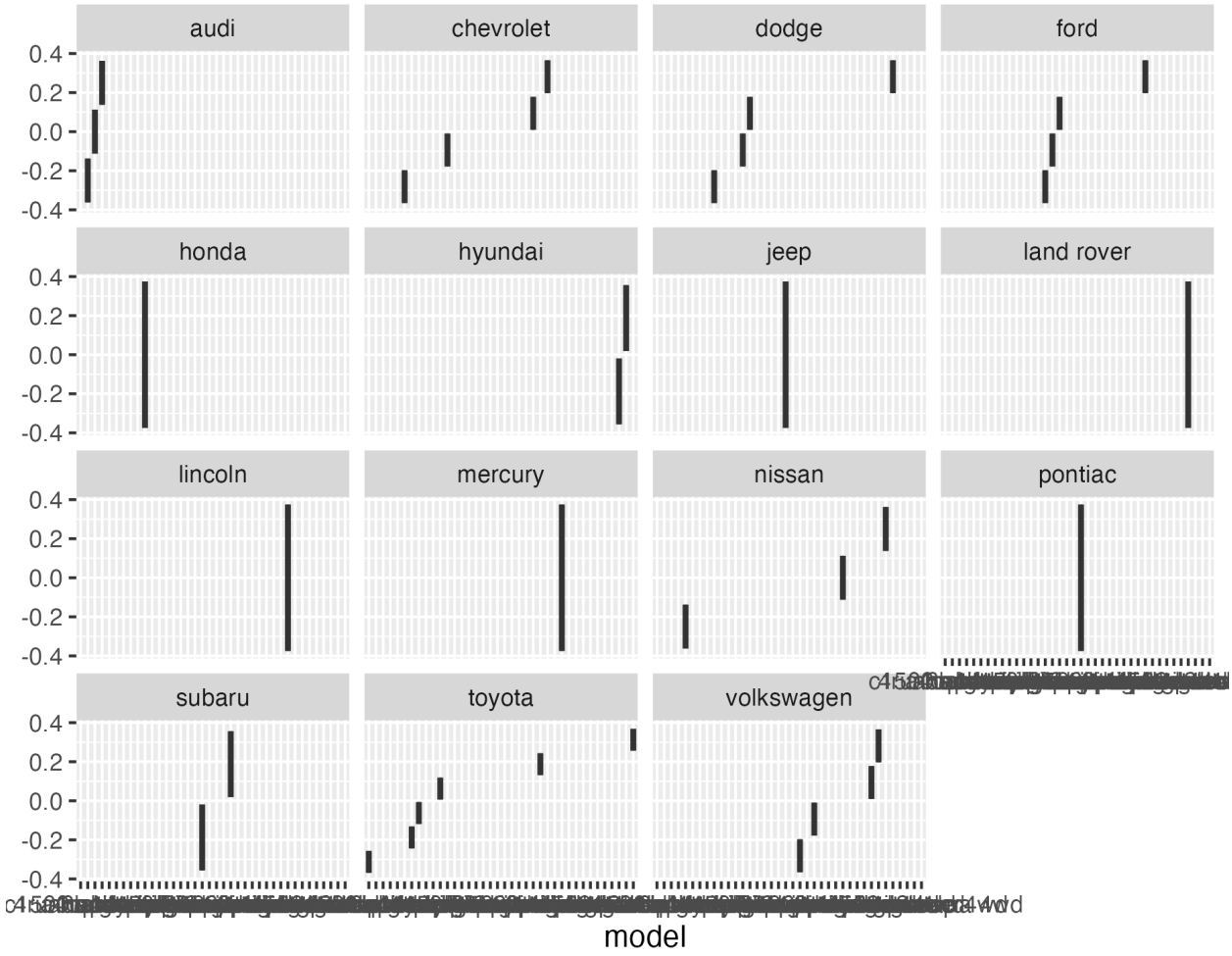
ggplot(mpg, aes(trans, class)) + geom_tile()



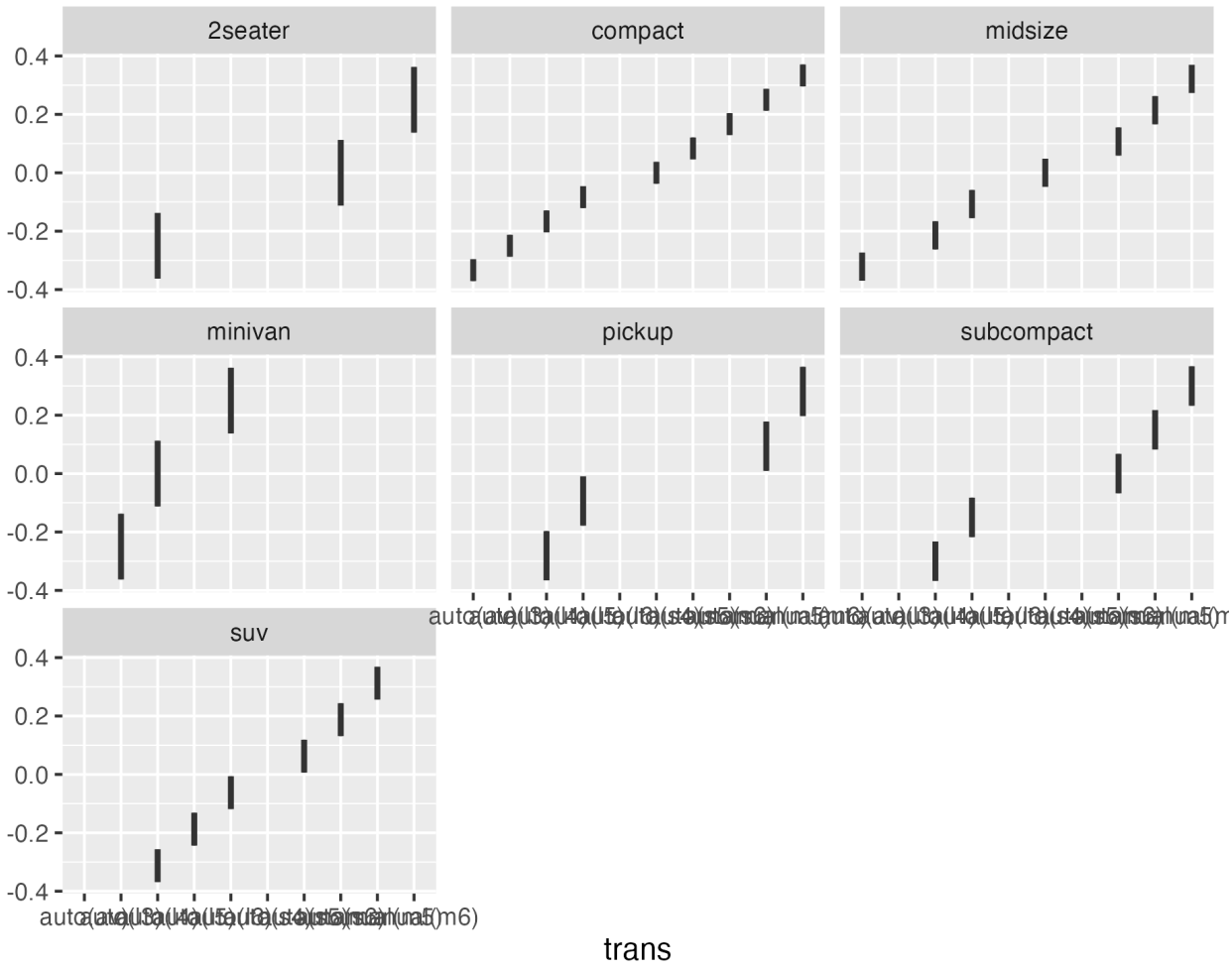
ggplot(mpg, aes(cyl, trans)) + geom_tile()



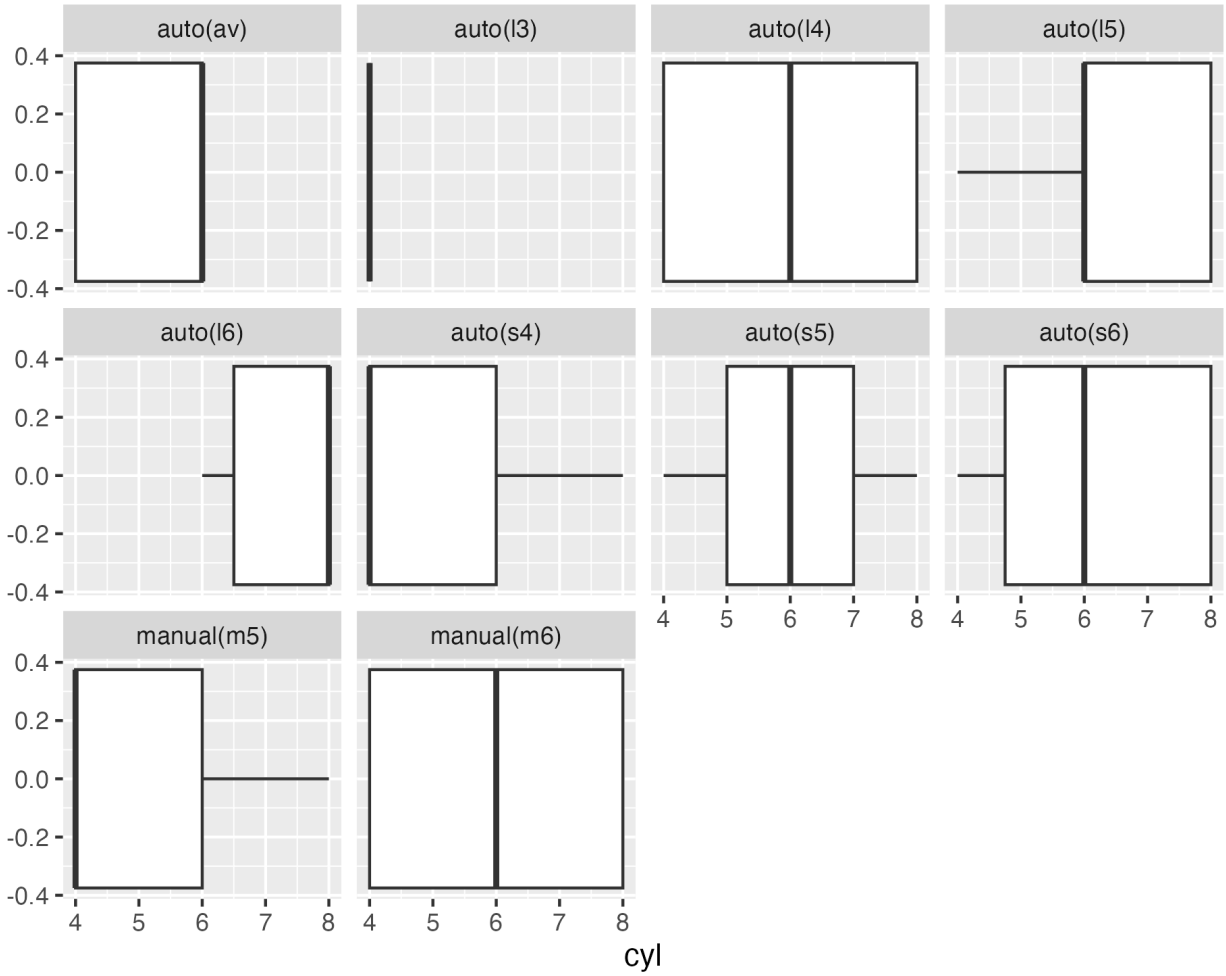
ggplot(mpg, aes(model)) + geom_boxplot() + facet_wrap(~manufacturer) # can use faceting



ggplot(mpg, aes(trans)) + geom_boxplot() + facet_wrap(~class)



ggplot(mpg, aes(cyl)) + geom_boxplot() + facet_wrap(~trans)


auto(av)
auto(l3)
auto(l4)
auto(l5)
auto(l6)
auto(s4)
auto(s5)
auto(s6)
manual(m5)
manual(m6)
0.4
0.2
0.0
-0.2
-0.4
4
5
6
7
8
cyl